

Highlights

Towards Smarter Customer Retention: A Review of Profit-Driven Churn Prediction including Fairness

Asif Iqbal, Juairia Moumie, Fowzia Bintu Karim, Jayanto Dutta Plabon, M. F. Mridha

- Summarizes the traditional machine learning models, ensemble models, deep learning and federated learning models for predicting customer churn.
- Talks about customer lifetime value, and profit-based evaluation as well as predictive accuracy for making profit-maximizing choices about customer retention.
- Identifies research gaps on the integration of profitability, explainability, fairness and privacy in proactive and personalized customer retention.

Towards Smarter Customer Retention: A Review of Profit-Driven Churn Prediction including Fairness

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ABSTRACT

Customer churn prediction is a critical task for improving customer retention and maximizing business profitability across customer-centric industries. Traditional machine learning techniques, including Logistic Regression, Decision Trees, and Support Vector Machines, have been widely applied for churn prediction; however, they often struggle to capture complex customer behavior in large-scale and high-dimensional datasets. Recent advances in ensemble learning and deep learning have significantly improved predictive performance, while business-oriented evaluation metrics such as Customer Lifetime Value (CLV), Expected Maximum Profit (EMPC), and e-Profits have shifted the focus from prediction accuracy to business value. Moreover, Explainable Artificial Intelligence (XAI) techniques, including SHAP and LIME, have enhanced model transparency, whereas fairness-aware approaches have been introduced to reduce algorithmic bias and ensure equitable decision-making. With increasing concerns regarding data privacy, Federated Learning (FL) has emerged as a promising paradigm that enables collaborative model training without sharing raw customer data. Furthermore, proactive churn management strategies leveraging real-time analytics and personalized retention have gained considerable attention. Despite these advancements, existing studies largely address these challenges independently, and a comprehensive framework integrating predictive performance, profitability, fairness, interpretability, and privacy remains limited. This review provides a comprehensive analysis of recent customer churn prediction techniques, discusses their strengths and limitations, and identifies key research gaps and future directions toward developing more effective, privacy-preserving, and business-oriented churn prediction systems.

1. Introduction

In years customer churn prediction has become a very important topic in business analytics because it directly affects how much money a company makes and how long it can stay in business. Keeping customers who already use a service is much cheaper than finding ones. This has made companies try to use ways to predict which customers might leave early Ouf, Mahmoud and Abdel-Fattah (2024). With more digital services and online platforms companies now have a lot of information about how customers behave what they buy and who they are. This helps them build models to predict churn.

In the past simple methods like regression and decision trees were used for this task because they are easy to understand. These methods often have trouble with complicated patterns in data Chen (2025). To do better companies started using advanced methods like Random Forest, Support Vector Machines and Gradient Boosting. These techniques work better in different areas like phone companies, demonstrating improved predictive performance across various domains including telecommunications, banking, and e-commerce Zhu (2024); Adeniran, Efunniyi, Osundare and Abhulimen (2024).

In years deep learning has taken things even further. It helps find patterns in data that has a lot of information or changes over time. Artificial Neural Networks and Long Short-Term Memory networks are being used more and more to predict customer churn because they can learn non-linear and sequential patterns, from customer data Khattak, Mehak, Ahmad et al. (2023); Equihua, Nordmark, Ali and Lausen (2023). These models are harder to understand which makes it harder for companies to use them in real life where people need to know why a decision is made Sikri, Jameel, Idrees and Kaur (2024).

To solve this some people have started using learning. This means using different models together to make better predictions. Research has increasingly looked into learning, where several predictive models are put together to get better results. This approach often performs better. Is more stable, than using just one model alone Araz et al. (2025); Wahul, Kale and Kota (2023). Some new systems, like those based on AutoGluon help pick the models and set their settings automatically Shoaib, Rahman, Maua, Rahman, Mridha, Kim and Shin (2025).

Some recent work shows that the usual ways to check how good a model is, like accuracy or AUC don't always tell the whole story. A new way to measure success called e-Profits looks at how money a model can actually make by considering things like how much a customer is worth over time and how much it costs to keep them Manzoor, Qureshi, Kidney and Longo (2026). This shows that a model that is not exactly perfect can still make money for a company.

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Another challenge is keeping customer data safe. Traditional centralized models need customer information to be gathered or processed at a central spot, which can raise privacy and security risks. Federated Learning offers a path by letting models be trained over many decentralized data sources without directly sharing the raw customer data McMahan, Moore, Ramage, Hampson and Aguera y Arcas (2017); Coimbra, Santos, Maia, Silva, Souza, Silva and Braga Silva (2024).

Also newer systems are getting better at making decisions and in real time. Of waiting for problems to happen companies now use AI to create special plans to keep each customer happy. Researchers are looking at methods such, as reinforcement learning and explainable artificial intelligence to help make retention decisions that're more personalized and transparent Mnih, Kavukcuoglu, Silver et al. (2015); Ip (2025); Asif, Arif and Mukheimer (2025).

With all these improvements many challenges still remain. I notice challenges such as class imbalance, model interpretability, algorithmic fairness, privacy and changes in customer behaviour, over time Mehrabi, Morstatter, Saxena, Lerman and Galstyan (2021); Guidotti, Monreale, Ruggieri, Turini, Giannotti and Pedreschi (2019); Suresh and Guttag (2021). Also putting together all these ideas into one system is still a challenge.

In this paper we look at all the ways people try to predict customer churn. We focus on machine learning, deep learning and new mixed methods. The main things we do in this study are:

- We look at all the machine learning and deep learning models that are used to predict customer churn.
- We study how money-focused ways to check models affect business choices.
- We look at methods that keep customer data safe like learning.
- We find the problems that are still not solved in these methods.
- We talk about what could be done in the future like AI, fast prediction systems and new ways to build models.

The rest of this paper is organized like this. Section 2 gives details.analysis of the customer churn datasets used in existing research, covering commonly used datasets from telecommunications, banking, retail, small and medium enterprises (SMEs), SaaS and OTT platforms, and other customer-centric domains. The section also discusses the characteristics, strengths, limitations, and suitability of different datasets for machine learning, deep learning, and profit-driven churn prediction.

Section 3 reviews the evolution of customer churn prediction from traditional Machine Learning (ML) approaches to Deep Learning (DL) and Federated Learning (FL), highlighting the development of Logistic Regression, Decision Trees, Random Forest, Support Vector Machines, ensemble

learning, neural networks, sequential models, and privacy-preserving distributed learning techniques.

Section 4 presents profit-aware customer churn prediction with Federated Learning, discussing traditional machine learning, ensemble learning, Automated Machine Learning (AutoML), deep learning, federated learning, and the limitations of existing approaches, followed by the identified literature gaps and the motivation for an integrated framework.

Section 5 discusses evaluation metrics and profit-driven customer churn prediction, beginning with traditional classification measures such as Accuracy, Precision, Recall, F1-score, and ROC-AUC, followed by evaluation under class imbalance, profit-oriented evaluation, Customer Lifetime Value (CLV), and individualized profit-based decision-making.

Section 6 focuses on Trustworthy Artificial Intelligence for customer churn prediction, covering Explainable Artificial Intelligence (XAI), SHAP and LIME-based explanations, algorithmic bias, fairness metrics, fairness-aware learning, and the balance among profitability, explainability, and fairness.

Section 7 presents proactive customer retention and prescriptive analytics, emphasizing the transition from predicting churn probability to recommending effective retention actions. This section discusses predictive-to-prescriptive analytics, reinforcement learning, personalized retention strategies, real-time churn analytics, and the challenges associated with proactive retention.

Section 8 provides a cross-domain analysis of customer churn prediction across telecommunications, banking and financial services, SMEs, SaaS and OTT platforms, e-commerce, healthcare, and insurance, highlighting domain-specific features, techniques, and challenges.

Section 9 presents a comparative analysis of representative customer churn prediction studies, examining their application domains, datasets, features, modeling techniques, and limitations.

Section 10 discusses the major research challenges and future directions, including overemphasis on predictive accuracy, behavioral changes and concept drift, limited cross-domain generalization, causal decision-making, explainability, fairness, privacy, advanced representation learning, multimodal customer intelligence, reinforcement learning, and autonomous intelligent retention ecosystems.

Finally, Section 11 concludes the paper by summarizing the major findings of the review and outlining future directions toward integrated, profitable, explainable, fair, privacy-preserving, and intelligent customer retention systems.

2. In-Depth Analysis of Customer Churn Datasets: Supporting Machine Learning and Profit-Driven Modeling

Customer churn prediction depends a lot on having data sets that show what customers do how they spend money and how they use services. These data sets are very important for building, checking and comparing machine learning and

deep learning models. They also help people who study this area understand problems like when some groups are much bigger than others when features are different in many ways and when there is not enough information.

Recently people have been doing research on how to make money and keep information safe. They use methods like Federated Learning. These methods require information that's unique and comes from different places. Customer churn data sets are useful because they help people understand how good models are at predicting things in the world.

This section is about some of the customer churn data sets that people use the most, for customer churn prediction. It is helpful to look at customer churn data sets because customer churn is an issue and people want to know how to predict customer churn.

2.1. IBM Telco Customer Churn Dataset

The IBM Telco Customer Churn dataset IBM (2018) is one of the datasets that people use a lot to figure out when customers stop using a service. This dataset has information about than 7,000 customers. The IBM Telco Customer Churn dataset includes things, like how customers stayed with the service what kind of contract they had, how much they paid every month how they paid their bills and what services they used from the IBM Telco Customer Churn dataset. The IBM Telco Customer Churn dataset also has a label that says if a customer left the service or not. The IBM Telco Customer Churn dataset is used a lot for training and checking machine learning models because it has a structure and the features are balance

2.2. Maven Telecom Churn Dataset

The Maven Telecom dataset Maven Analytics (2020) has a lot of information about telecom customers. This information includes how telecom customers used the services of Maven Telecom, how they paid for these services and who the telecom customers are. The Maven Telecom dataset is very helpful, for looking at how the telecom customers are divided into different groups and how the telecom customers act. The Maven Telecom dataset works well for both classifying the telecom customers and making money from the telecom customers, which makes the Maven Telecom dataset good for projects that predict when telecom customers will stop using the services of Maven Telecom.

2.3. Bank Marketing Dataset

The Bank Marketing is a dataset with customer and campaign information to predict the probability of a customer subscribing to a term deposit Moro, Cortez and Rita (2014). It is applicable to customer-response modelling and targeted marketing but the response variable is not churn but subscription.

2.4. Credit Card Customer Dataset

The Credit Card dataset Shoaib et al. (2025) has information, about customers and their accounts. It shows how much they can spend, what they purchase and how they make payments. This dataset is used often for predicting customer

loss in the sector. The dataset allows researchers to study loyalty and spending patterns which're essential for retaining customers.

2.5. Retail Banking Churn Dataset

Retail banking dataset Prakash (2024) have customer account details how they. What services they use. These datasets are used to predict if customers leave banking services. They help show what makes customers happy how stable they are financially and what services they like which are important for losing customers.

2.6. Telecom Big Data Datasets

Large telecom dataset Ouf et al. (2024) have a lot of customer information. This includes phone calls, internet use, subscriptions and other service interactions. These dataset are often not balanced and need ways to prepare them. They are important for building models that work well in situations.

2.7. SME Customer Dataset

Small and Medium Enterprise datasets Ahmad and Islam (2021) examine how to use machine learning to forecast customer churn, in Medium Enterprises. I think these studies matter because Small and Medium Enterprises often have customer traits and operational needs that differ from those of large organizations.

2.8. SaaS and OTT Customer Dataset

The SaaS and OTT dataset Devi, Pothini, Kumari, V and Charan (2023) has information on how people subscribe to things and how they use these services. The SaaS and OTT dataset is really good at showing us how things change over time. This is useful for things like LSTM models that look at patterns that happen over time. The SaaS and OTT dataset is great for this because it has a lot of information, on how people use the SaaS and OTT services over time.

2.9. MultiDomain Customer Datasets

Recent studies Shoaib et al. (2025) use dataset from areas like telecom, banking, health care and online shopping. This is really helpful because it makes the models work better in places. The data from telecom, banking, health care and online shopping helps make models that work the way in these industries. For example models that are trained on data from telecom can also work well in banking and health care and online shopping. So the data from telecom, banking, health care and online shopping is very useful, for making models that work well in all these areas.

3. Evolution from Machine Learning to Federated Learning for Customer Churn Prediction

Customer churn prediction has changed a lot over the ten years because of fast changes in Artificial Intelligence, Machine Learning, Deep Learning and privacy-focused ways to work together. At first old ways of looking at data and

Table 1
Commonly Used Customer Churn Datasets

Dataset	Features	Strengths	Limitations
IBM Telco Dataset IBM (2018)	Tenure charges, contracts, internet services, churn label	Benchmark dataset; easy preprocessing; suitable for ML/DL	Small scale; lacks temporal behavior data
Maven Telecom Dataset Maven Analytics (2020)	Billing usage patterns, demographics, subscriptions	Good for behavioral analysis and profit-driven modeling	Imbalanced data; limited documentation
Bank Marketing Dataset Moro et al. (2014)	Age job, campaign interactions, previous outcomes	Useful for classification and marketing analysis	Not specifically designed for churn prediction
Credit Card Dataset Kaggle (2019)	Transactions credit limits, payment history	Effective for financial churn analysis	Privacy concerns and preprocessing complexity
Retail Banking Dataset Prakash (2024)	Transactions account activity, service usage	Supports banking retention analysis	Mostly proprietary and non-standardized
Telecom Big Data Dataset Ouf et al. (2024)	Call records internet usage, complaints	Suitable for scalable churn prediction	Highly imbalanced and resource-intensive
SME Customer Dataset Ahmad and Islam (2021)	Business transactions and service usage	Useful for sparse-data environments	Limited data size and generalization
SaaS and OTT Dataset Devi et al. (2023)	Login activity, subscriptions support tickets	Suitable for LSTM and sequential analysis	Requires continuous re-training
Multi-Domain Datasets Shoaib et al. (2025)	Telecom, banking, health-care, e-commerce data	Improves cross-domain generalization	Complex preprocessing and integration

using machine learning were used to find customers who might stop using services. Deep learning helped make predictions better by learning patterns and time changes from how customers act. However as companies worried more, about keeping customer data safe and sharing data properly Federated Learning came up. This way lets models be trained in places without sharing the actual customer data Vafeiadis, Diamantaras, Sarigiannidis, Chatzisavvas and Vassilopoulos (2015); Verbeke, Martens, Mues and Baesens (2011); Imani, Joudaki, Beikmohammadi and Arabnia (2025); McMahan et al. (2017).

This change from machine learning to learning and then to federated learning shows a slow move, toward better bigger more accurate and more private ways to predict customer loss Imani et al. (2025); McMahan et al. (2017).

3.1. Machine Learning Approaches for Customer Churn Prediction

Traditional Machine Learning techniques laid the foundation for modern customer churn prediction systems. Early studies mainly focused on supervised learning algorithms that classify customers into churners and non-churners based on historical customer information such as demographics, billing records, subscription history, customer complaints,

transaction frequency, and service usage patterns Vafeiadis et al. (2015); Verbeke et al. (2011).

Among the most widely used ML algorithms are Logistic Regression (LR), Decision Trees (DT), Random Forest (RF), Support Vector Machines (SVM), Naive Bayes (NB), and Gradient Boosting methods such as XGBoost and LightGBM. These models became popular because of their simplicity, interpretability, and computational efficiency Vafeiadis et al. (2015); Wahul et al. (2023); Erickson, Mueller, Shirkov, Zhang, Larroy, Li and Smola (2020); Imani et al. (2025).

Logistic Regression is one of the earliest and most widely applied techniques for churn prediction. It estimates the probability of churn using the sigmoid activation function Vafeiadis et al. (2015):

$$P(y = 1|X) = \frac{1}{1 + e^{-(\beta_0 + \beta^T X)}} \quad (1)$$

where:

- X represents the customer feature vector,
- β denotes model coefficients,
- $P(y = 1|X)$ indicates the probability of customer churn.

Decision Tree and Random Forest models further improved churn classification by capturing nonlinear relationships among customer attributes. Random Forest combines multiple decision trees to reduce overfitting and improve prediction robustness. The prediction function of Random Forest can be represented as Verbeke et al. (2011); Wahul et al. (2023):

$$\hat{f}(x) = \frac{1}{B} \sum_{b=1}^B f_b(x) \quad (2)$$

where $f_b(x)$ denotes the prediction from the b -th decision tree.

Similarly, Support Vector Machines maximize the margin between churn and non-churn classes to achieve accurate classification. Ensemble learning algorithms such as XG-Boost and LightGBM later became highly effective because they sequentially optimize weak learners and minimize prediction errors Vafeiadis et al. (2015); Erickson et al. (2020); Imani et al. (2025).

Traditional ML models provide several advantages:

- High interpretability,
- Fast training and inference,
- Lower computational complexity,
- Easy deployment in industrial systems.

Despite these benefits, traditional ML approaches face several limitations. Most models depend heavily on manual feature engineering and struggle to capture complex nonlinear customer behaviors and temporal dependencies. Moreover, ML models are often less effective when dealing with high-dimensional behavioral datasets and sequential interaction data Imani et al. (2025).

3.2. Deep Learning-Based Customer Churn Prediction

To overcome the limitations of traditional ML methods, Deep Learning methods have drawn a lot of interest in churn prediction studies because they can automatically discover patterns and complicated nonlinear connections, in customer information Khattak et al. (2023); Equihua et al. (2023); Imani et al. (2025).

Artificial Neural Networks, Recurrent Neural Networks, Convolutional Neural Networks and Long Short-Term Memory networks are, among the architectures explored for customer churn analytics Khattak et al. (2023); De Caigny, Coussement, De Bock and Lessmann (2020); Equihua et al. (2023).

ANN models consist of multiple hidden layers that transform input customer features into higher-level representations. The output of a neural network layer is computed as:

$$a_j = \sigma \left(\sum_{i=1}^n w_{ij} x_i + b_j \right) \quad (3)$$

where:

- x_i denotes input features,
- w_{ij} represents connection weights,
- b_j is the bias term,
- $\sigma(\cdot)$ is the activation function.

Deep Learning models became particularly important because customer churn behavior often evolves over time. Sequential customer interactions such as browsing activity, payment history, subscription renewals, complaint records, and application usage patterns contain temporal dependencies that traditional ML methods fail to capture efficiently Equihua et al. (2023); Imani et al. (2025).

LSTM networks address this challenge through memory cells and gating mechanisms that preserve long-term dependencies. The forget gate in LSTM is defined as Devi et al. (2023); Equihua et al. (2023):

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \quad (4)$$

where:

- h_{t-1} is the previous hidden state,
- x_t represents the current input,
- W_f and b_f denote trainable parameters.

Deep Learning techniques significantly improved churn prediction performance because they:

- Learn complex nonlinear customer relationships,
- Capture sequential and temporal customer behavior,
- Reduce dependency on manual feature engineering,
- Handle high-dimensional behavioral datasets effectively.

However, DL models also introduce several challenges:

- High computational and memory requirements,
- Need for large-scale labeled datasets,
- Long training times,
- Reduced interpretability compared to traditional ML models.

Table 2
Traditional Machine Learning Algorithms for Customer Churn Prediction

Algorithm	Learning Type	Advantages	Limitations
Logistic Regression Vafeiadis et al. (2015)	Supervised Learning	Simple, interpretable, computationally efficient	Limited capability for nonlinear feature interactions
Decision Tree Verbeke et al. (2011)	Supervised Learning	Easy visualization and rule extraction	Prone to overfitting and instability
Random Forest Wahul et al. (2023)	Ensemble Learning	High accuracy and robustness against overfitting	Increased computational complexity
Support Vector Machine (SVM) Vafeiadis et al. (2015)	Supervised Learning	Effective in high-dimensional spaces	Slow training on large-scale datasets
XGBoost Erickson et al. (2020); Imani et al. (2025)	Gradient Boosting	Excellent predictive performance and scalability	Requires hyperparameter tuning
LightGBM Erickson et al. (2020); Imani et al. (2025)	Gradient Boosting	Faster training and lower memory consumption	Sensitive to noisy and imbalanced data

Table 3
Deep Learning Models for Customer Churn Prediction

Model	Primary Application	Strengths	Challenges
Artificial Neural Network (ANN) Khattak et al. (2023)	General churn classification	Learns complex nonlinear relationships	Requires large training data
Recurrent Neural Network (RNN) Equihua et al. (2023)	Sequential customer behavior analysis	Handles temporal dependencies	Suffers from vanishing gradient problem
Long Short-Term Memory (LSTM) Equihua et al. (2023)	Time-series churn prediction	Captures long-term customer behavior patterns	Computationally expensive
Convolutional Neural Network (CNN) De Caigny et al. (2020)	Feature extraction from behavioral patterns	Learns local feature interactions effectively	Less interpretable than traditional ML models
Hybrid DL Models Khattak et al. (2023); Imani et al. (2025)	Multi-source customer analytics	Improves overall prediction performance	Complex architecture design

3.3. Federated Learning for Privacy-Preserving Customer Churn Prediction

Although ML and DL techniques substantially improved churn prediction accuracy, both approaches commonly rely on centralized training architectures where customer data from multiple organizations is collected into a single server.

This centralized paradigm creates serious concerns regarding data privacy, regulatory compliance, communication overhead, and cybersecurity risks McMahan et al. (2017); Imani et al. (2025).

Industries such as telecommunications, banking, healthcare, insurance, and subscription-based services manage

highly sensitive customer information. Sharing customer-level data among organizations often violates privacy regulations such as the General Data Protection Regulation (GDPR). Consequently, Federated Learning (FL) emerged as a revolutionary distributed learning framework designed to preserve privacy while enabling collaborative model training McMahan et al. (2017).

Federated Learning allows multiple organizations to train a shared churn prediction model without exchanging raw customer data. Instead, each organization trains the model locally and sends only model parameters or gradients to a central aggregation server McMahan et al. (2017).

Let the set of participating organizations be represented as:

$$\Pi = \{1, 2, 3, \dots, \mu\} \quad (5)$$

where each participant $\pi \in \Pi$ possesses a private dataset Δ_π .

The local churn prediction objective function is expressed as:

$$\Phi_\pi(\omega_\pi) = -\frac{1}{N} \sum_{j=1}^N [y_j \log(\hat{y}_j) + (1 - y_j) \log(1 - \hat{y}_j)] \quad (6)$$

The global federated optimization objective is:

$$\Omega_G^* = \arg \min \sum_{\pi=1}^{\mu} \frac{|\Delta_\pi|}{|\Delta|} \Phi_\pi(\omega_\pi) \quad (7)$$

The Federated Averaging (FedAvg) aggregation mechanism is defined as McMahan et al. (2017):

$$\Omega_G^{\tau+1} = \sum_{\pi \in S^\tau} \frac{|\Delta_\pi|}{|\Delta|} \omega_\pi^{\tau+1} \quad (8)$$

where:

- S^τ represents selected participants,
- $\omega_\pi^{\tau+1}$ denotes updated local model parameters.

Federated Learning offers several important benefits:

- Strong privacy preservation,
- Regulatory compliance support,
- Collaborative cross-organization intelligence,
- Scalability across distributed environments,
- Improved generalization using diverse customer behaviors.

Despite these advantages, FL also faces several research challenges including:

- Communication overhead,
- Non-IID customer data distribution,
- Client drift and convergence instability,
- Security threats such as poisoning attacks,
- System heterogeneity.

The goal of federated optimization methods is to cope with the problems due to the heterogeneous data and computing resources at the client side. FedProx increases the robustness of the algorithm under statistical and system heterogeneity Li, Sahu, Zaheer, Sanjabi, Talwalkar and Smith (2020) and SCAFFOLD is used to decrease client drift during training by using control variates Karimireddy, Kale, Mohri, Reddi, Stich and Suresh (2020).

4. Profit-Aware Customer Churn Prediction with Federated Learning

Customer churn prediction has become one of the most extensively studied research areas in Customer Relationship Management (CRM) because customer retention is considerably more cost-effective than customer acquisition. Organizations continuously invest in predictive analytics to identify customers who are likely to discontinue their services so that proactive retention strategies can be implemented before customer defection occurs Ngai, Xiu and Chau (2009); Hadden, Tiwari, Roy and Ruta (2007). Numerous studies have demonstrated that even a small improvement in customer retention can substantially increase long-term organizational profitability due to the recurring value generated by loyal customers Lemmens and Gupta (2020); Gupta, Lehmann and Stuart (2004).

With the rapid advancement of Artificial Intelligence (AI), Machine Learning (ML), and data-driven business analytics, customer churn prediction has evolved from traditional statistical analysis to sophisticated intelligent decision-support systems. Modern churn prediction models are capable of analyzing customer demographics, service usage patterns, billing information, complaints, transaction history, and behavioral characteristics to estimate the probability of customer attrition with high accuracy Barsotti et al. (2024); Imani et al. (2025). These predictive systems enable organizations to design personalized marketing campaigns, optimize customer retention costs, and improve overall business performance.

Researchers have proposed numerous approaches for churn prediction during the last two decades. As illustrated in Figure 1, the evolution of churn prediction techniques can generally be categorized into the following stages:

- Traditional statistical and machine learning approaches.
- Ensemble learning algorithms.
- Deep learning models.
- Explainable Artificial Intelligence (XAI).

Evolution of Customer Churn Prediction

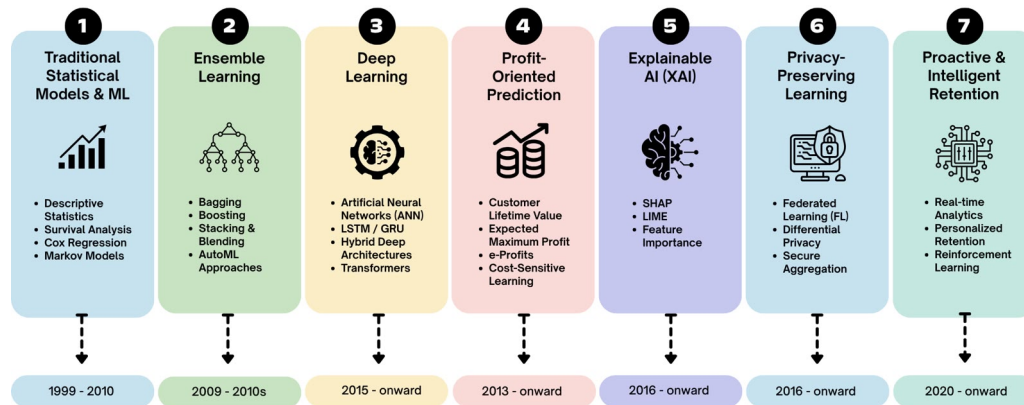


Figure 1: Evolution of Customer Churn Prediction

- Profit-driven and fairness-aware customer analytics.
- Privacy-preserving Federated Learning.
- Proactive and intelligent retention.

Each generation has improved different aspects of churn prediction, including predictive accuracy, interpretability, scalability, privacy preservation, and business profitability. The following subsections discuss these developments in detail.

4.1. Traditional Machine Learning Approaches

Traditional machine learning algorithms have served as the foundation of customer churn prediction for many years. Early studies mainly relied on classification algorithms because churn prediction is naturally formulated as a binary classification problem in which customers are categorized as either churners or non-churners Verbeke et al. (2011); Coussement and Van den Poel (2008). These models are not hard to work with easy to put into action and give lines of decision which makes them good for use in factories and companies.

Logistic Regression is one of the methods used for predicting customers leaving because it is simple and gives clear chances. It figures out the chance a customer will leave by looking at customer details and how likely they are to leave. Even though Logistic Regression is clear about how it makes decisions it does not work well when the way customers act is very complex and not straight Vafeiadis et al. (2015).

Decision Tree methods came later because they can handle patterns and give rules that people can understand. Their way of organizing information helps people who work with data to see what is most important in customers leaving.. Decision Trees can be affected by bad data and often get too specific to the data they are trained on Verbeke et al. (2011); Hadden et al. (2007).

Support Vector Machines (SVM) are also used a lot for predicting customers leaving because they can draw the lines between different groups in a lot of details. SVM usually

does better than older methods especially if the right ways to look at the data are used. However it takes time and work when the amount of data is very big which makes it harder to use in real situations Coussement and Van den Poel (2008); Vafeiadis et al. (2015).

Random Forest made things better by using Decision Trees together. This way the results are more accurate and easier to rely on. Random Forest also helps find which details matter most for customers leaving. Many studies show that Random Forest does better than one single Decision Tree and is better at handling messy data Chen (2025); Mettle, Ayerko and Appiahene (2026).

Many studies have checked how well old machine learning methods work on data about customers leaving. Even though no one method is always the best, Decision Trees, Random Forest Support Vector Machines and Logistic Regression are still used a lot because they are not too hard to use and are easy to understand Barsotti et al. (2024); Imani et al. (2025).

The main good parts of machine learning methods are:

- They are not hard to use and do not need computing power.
- It is easy to understand what the results mean.
- They train quickly on data that is organized in tables.
- They are good to use as a starting point to check methods.

But these older methods also have some big problems:

- They can't handle when customer behavior is very complicated.
- They do not work well when the data has a lot of different details.
- They are affected by when some types of data are more common than others.

- They do not work well when the amount of data grows fast.

These problems made people look for better ways to make models that can do better and still be strong. So complex group methods like XGBoost, LightGBM and CatBoost became the main ways to predict customers leaving.

4.2. Ensemble Learning Approaches

The problems with machine learning methods made people create group methods that use many small models together to make a better one. Group methods improve how well things are predicted by making the results more stable not making the model too good for the data and understanding patterns in the data Wahul et al. (2023); Araz et al. (2025).

In group methods boosting is one of the ways for data about customers. Of making models separately boosting methods add many decision trees one after another. Each new tree tries to fix the mistakes from the ones. This way the results get better. The model works well with new data.

XGBoost has become one of the used ways to predict customers leaving because it works fast helps avoid making the model too good for the data and can handle missing information on its own. Many studies in telecom and finance show that XGBoost is better than ways like Logistic Regression, Decision Trees and Support Vector Machines Zhu (2024); Sikri et al. (2024).

Two different metrics have been used to evaluate LightGBM for customer churn prediction: predictive metrics and profit-based metrics. It may rank better than its traditional ranking measures, emphasizing the need for assessing models based on business goals. Manzoor et al. (2026).

In the telecom churn prediction task, CatBoost has also been tested. Based on the results, CatBoost can outperform the conventional accuracy/AUC ranking measures in the profit-based evaluation even if it is not the top-performing model in the conventional rank-based evaluation methods Manzoor et al. (2026).

Recent tests say that no one group method is always the best. How well a method works depends on what the data's like how good the details are and how the method is set up. So many people suggest trying group methods before choosing the one to use Barsotti et al. (2024); Imani et al. (2025).

The main good parts of group methods are:

- They are better at predicting than models.
- They are better, at handling customer data.
- Reduced overfitting through model aggregation.
- Strong capability to capture nonlinear customer behavior.
- Excellent performance on structured tabular datasets.

Despite these advantages, ensemble models generally require careful hyperparameter tuning and increased computational resources during training. Moreover, most ensemble methods operate as black-box models, making their predictions difficult for business analysts to interpret directly.

4.3. Automated Machine Learning (AutoML)

The increasing complexity of machine learning model selection has encouraged the development of Automated Machine Learning (AutoML) systems. AutoML frameworks automatically perform data preprocessing, feature engineering, algorithm selection, hyperparameter optimization, and ensemble generation with minimal human intervention.

Among existing AutoML frameworks, AutoGluon has attracted considerable attention because of its outstanding performance on structured tabular datasets. AutoGluon automatically constructs multiple machine learning models and combines them into weighted ensembles to improve prediction accuracy Erickson et al. (2020).

Recent studies demonstrate that AutoGluon ensembles not only improve predictive accuracy but also enhance business profitability through automated ensemble optimization. Such frameworks significantly reduce manual experimentation while maintaining state-of-the-art predictive performance across multiple business applications Shoaib et al. (2025); Araz et al. (2025).

Although AutoML simplifies model development, its computational cost may be higher than manually optimized models because multiple candidate models are trained simultaneously. Furthermore, the automatically generated ensemble architecture may be difficult to interpret without additional explainability techniques.

4.4. Deep Learning Approaches

Deep Learning has emerged as another promising direction for customer churn prediction due to its ability to automatically learn hierarchical feature representations from large-scale datasets. Unlike traditional machine learning algorithms that rely heavily on manual feature engineering, ... deep neural networks learn complex nonlinear relationships from customer data Khattak et al. (2023); Imani et al. (2025).

Deep Neural Networks DNNs have shown they can learn levels of features in predicting customer churn Khattak et al. (2023); Devi et al. (2023).

Convolutional Neural Networks (CNNs), originally developed for computer vision tasks, have also been successfully adapted for customer churn prediction. CNN-based models automatically identify local feature interactions and have shown competitive performance for structured customer datasets when combined with feature transformation techniques De Caigny et al. (2020); Equihua et al. (2023).

Recurrent Neural Networks and Long Short-Term Memory networks can model customer behaviour. Recurrent Neural Networks and Long Short-Term Memory networks also capture temporal dependencies Equihua et al. (2023); Devi et al. (2023).

More recently, Transformer-based architectures have demonstrated outstanding performance in learning long-range dependencies from sequential customer behavior. Their self-attention mechanism enables efficient modeling of complex customer interactions across long observation periods while supporting parallel computation during training Wu et al. (2020); Chen, Lv, Wang, Xiang, Wu, Luo and Zhang (2025).

Despite their impressive predictive capability, deep learning models require considerably larger datasets, higher computational resources, and longer training time than conventional machine learning algorithms. Furthermore, their black-box nature reduces model transparency, making it difficult for business analysts to understand why a customer is predicted to churn. These limitations have motivated the integration of Explainable Artificial Intelligence (XAI) techniques into deep learning-based customer churn prediction systems Guidotti et al. (2019); Arrieta, Diaz-Rodriguez, Del Ser, Bennetot, Tabik, Barbado et al. (2020).

4.5. Federated Learning for Customer Churn Prediction

The rapid growth of customer data has significantly improved the capability of machine learning models to predict customer churn. However, most existing churn prediction systems rely on centralized learning, where customer data from multiple organizations are collected into a single server before model training. Although this approach generally achieves high predictive performance, it introduces serious concerns regarding customer privacy, data security, regulatory compliance, and organizational data ownership McMahan et al. (2017); Rieke, Hancox, Li, Milletari, Roth, Albarqouni et al. (2020).

The increasing adoption of privacy regulations such as the General Data Protection Regulation (GDPR) and the California Consumer Privacy Act (CCPA) has further limited the direct sharing of customer information across organizations. Consequently, many companies are unable to collaborate even when joint learning could substantially improve predictive performance. These challenges have motivated researchers to investigate Privacy-preserving machine learning techniques that allow for collaborative knowledge sharing, without revealing customer information McMahan et al. (2017); Coimbra et al. (2024).

Federated Learning has become one of the promising ways to handle these kinds of problems. Of sending customer data to a central place each client works on its own private data to train a machine learning model. Only the model changes or model parameters are sent to a server. There all these changes are put together to create an overall model. Because customer data stays on devices or within the organization Federated Learning helps keep things private and still allows people to work together McMahan et al. (2017); Coimbra et al. (2024).

One of the important ways to optimize Federated Learning is called Federated Averaging. This method was created by McMahan and others. It takes the model changes from

each client. Combines them using a weighted average based on how much data each client has. This simple way of working helps save on communication costs and still keeps customer data McMahan et al. (2017). Because it is easy to use and works well Federated Averaging is the starting point for research in Federated Learning.

Even though Federated Averaging works well when all the clients have data it is not as good when the data is different between clients. In situations like predicting when customers leave different organizations have different kinds of customers different services, different places and different ways of doing business. This kind of data called Non-IID often makes it hard for the model to get better. Can make predictions worse Li et al. (2020); Karimireddy et al. (2020).

To solve these problems some new methods have been created. One is called FedProx. It adds a part to Federated Averaging that stops the local models from changing too much. This helps keep the models stable when the client data is different Li et al. (2020). Another method is FedDyn, which uses a changing way to keep the model updates under control. This helps stop models from moving far away from the main model during training Acar, Zhao, Matas, Mattina, Whatmough and Saligrama (2021).

Another big improvement is SCAFFOLD. This method uses tricks to fix problems that happen when the data between clients is different. By keeping the models more in line with the main model SCAFFOLD helps the model get better faster and be more accurate, than the older FedAvg method when the data is not the same Karimireddy et al. (2020). Adaptive Federated Optimization methods further improve server-side optimization by incorporating adaptive learning rates, resulting in better convergence stability and communication efficiency Reddi, Charles, Zaheer, Garrett, Rush, Konecný, Kumar and McMahan (2021).

Recent studies demonstrate that Federated Learning has been successfully applied in several privacy-sensitive domains, including privacy-sensitive domains, including healthcare and other distributed learning environments Rieke et al. (2020); McMahan et al. (2017). Consequently, Federated Learning has become an attractive solution for customer churn prediction where organizations are unwilling or legally unable to exchange raw customer information.

Despite these advances, several important challenges remain unresolved. Many federated learning systems assume the model structure, for all the clients taking part. This can be limiting when organizations have computing power business needs or data types McMahan et al. (2017); Li et al. (2020). For example, one telecom operator may utilize XGBoost, another may employ CatBoost, while another organization may prefer LightGBM or TabPFN. Conventional parameter averaging cannot directly aggregate such heterogeneous models.

Furthermore, The standard FedAvg method gives weight to clients that have more training data. However recent business research points out that profit and fairness can also matter when designing customer analytics systems McMahan et al. (2017); Rahman, Janssens and Bogaert (2025).

Several recent studies have emphasized that future Federated Learning research should move beyond conventional parameter averaging toward adaptive aggregation strategies capable of integrating heterogeneous learning models while simultaneously considering privacy preservation, business profitability, explainability, and fairness Barsotti et al. (2024); Imani et al. (2025).

The major advantages of Federated Learning can be summarized as follows:

- Preserves customer privacy by keeping raw data on local devices.
- Enables collaborative model training among multiple organizations.
- Reduces regulatory concerns associated with centralized data collection.
- Improves scalability for distributed enterprise environments.
- Supports decentralized artificial intelligence applications.

However, Federated Learning also exhibits several limitations:

- Performance degradation under Non-IID client distributions.
- Communication overhead during multiple training rounds.
- Difficulty handling heterogeneous machine learning models.
- Limited integration of explainability and fairness techniques.
- Most existing aggregation methods ignore business-oriented optimization objectives.

These limitations motivate the development of more advanced heterogeneous Federated Learning frameworks capable of integrating diverse machine learning models while simultaneously optimizing predictive accuracy, customer privacy, business profitability, fairness, and explainability. Such integrated frameworks represent an important future direction for intelligent customer churn prediction systems.

4.6. Summary of Literature and Research Gap

The reviewed literature demonstrates that customer churn prediction has evolved significantly over the past two decades. Earlier studies primarily focused on conventional machine learning algorithms such as Decision Trees, Logistic Regression, Random Forest, Support Vector Machine (SVM), and Gradient Boosting to classify churn-prone customers Verbeke et al. (2011); Vafeiadis et al. (2015). The ways we look at customer behavior have really helped us understand why customers stop doing business with us.

We can use what customers did in the past to figure out if they will leave us in the future.

Lately people have been working on ways to make predictions better. They are using things like XGBoost, LightGBM, CatBoost, AutoML frameworks and deep neural networks. These methods are really good at finding patterns in customer data. They are a lot better, than the ways of doing things Erickson et al. (2020); Araz et al. (2025).

In addition, modern studies increasingly emphasize customer lifetime value, profit maximization, and personalized retention strategies rather than relying solely on prediction accuracy Jiang, Liu, Abedin, Wang, Yang and Dong (2024); Gómez-Vargas, Maldonado and Vairetti (2025). People are also talking about privacy because customer information is shared between companies and there are laws to protect it.

Customer data is very important. We need to make sure it is safe. Federated Learning has emerged as a promising paradigm that enables collaborative model training without transferring raw customer data to a central server McMahan et al. (2017); Li et al. (2020). Several optimization strategies including FedProx, FedDyn, SCAFFOLD, and Adaptive Federated Optimization have further improved convergence under heterogeneous environments Acar et al. (2021); Karimireddy et al. (2020). However, most federated approaches focus primarily on optimization efficiency and predictive performance while overlooking business objectives.

Explainable Artificial Intelligence (XAI) has become another essential component of trustworthy churn prediction systems. Techniques such as SHAP and LIME provide transparent explanations for individual predictions and help business managers understand the contribution of each feature toward churn decisions Lundberg, Erion, Chen, DeGrave, Prutkin, Nair, Katz, Himmelfarb, Bansal and Lee (2020); Ribeiro, Singh and Guestrin (2016). Fairness-aware machine learning has also received increasing attention to reduce algorithmic bias against demographic groups and ensure ethical deployment of AI systems Mehrabi et al. (2021); Pes-sach and Shmueli (2022). Nevertheless, explainability and fairness are often evaluated independently instead of being integrated into a unified customer retention framework.

Profit-oriented churn prediction has gained considerable interest in recent years. Studies have proposed cost-sensitive learning, profit-driven evaluation metrics, customer lifetime value optimization, and business-oriented decision frameworks to maximize organizational revenue Verbraken, Verbeke and Baesens (2013); Höppner, Stripling, Baesens, Van den Broucke and Verdonck (2020). The recently proposed e-Profits metric further aligns model evaluation with real business profit instead of relying only on traditional classification measures Manzoor et al. (2026). Despite these developments, very few studies combine profit-aware learning with privacy-preserving federated architectures. Recent surveys show that current work on predicting customer churn is starting to focus on systems that combine different types of intelligence. These systems aim to include things like being easy to understand being fair making sure the business makes money and working across places.. Right now most of

the work looks at each of these parts on their own instead of creating a full system that works together Imani et al. (2025); Barsotti et al. (2024).

Because of this there is still a lot of room to create a system that handles all these things at the time. This system would need to be good, at predicting churn keep customer information be fair be easy to explain and also help the business make more money.

4.7. Research Gap

When we look at all the research that has been done we find some gaps that need to be filled.

- A lot of the studies that're already out there try to make predictions that are as accurate as possible but they do not think about how this will affect the business in terms of making money or how valuable customers are over time when they are making their models Lemmens and Gupta (2020); Gómez-Vargas et al. (2025).
- FedAvg averages the client updates by their local training-data sizes McMahan et al. (2017). This weighting rule does not explicitly include customer value or the profitability of retaining customers, thus the research into business-aware aggregation strategies.
- Using explainable churn models, features which affect the prediction are identified Maan and Maan (2023). An exciting future research direction is to link these explanations to retention-action selection and examine the business outcomes.
- Fairness is generally evaluated separately from prediction performance and rarely optimized together with profitability, making practical deployment challenging Mehrabi et al. (2021); Pessach and Shmueli (2022).
- Traditional classification measures are not direct measures of the financial consequences of retention decisions. A profit based evaluation can give a different ranking for the models than the traditional predictive evaluation, which means that business-oriented measures can be also incorporated in the churn evaluation Manzoor et al. (2026).
- Current research generally applies homogeneous learning models in federated environments, whereas real-world organizations often employ heterogeneous machine learning models with different learning capabilities Li et al. (2020); Acar et al. (2021).
- Very few studies integrate Customer Lifetime Value (CLTV), Explainable AI, Federated Learning, Fairness, and profit-sensitive evaluation into one unified customer churn prediction framework Rahman et al. (2025); Manzoor et al. (2026).

5. Evaluation Metrics and Profit-Driven Customer Churn Prediction

Evaluating customer churn prediction models has become increasingly important as research has evolved from simply identifying customers likely to leave toward supporting profitable and actionable retention strategies. Previous work focused more on prediction performance via traditional classification metrics, but more recent work has adopted a business-oriented evaluation, incorporating the customer lifetime value (CLV), retention cost, and expected financial return into the evaluation process. This evolution is due to the increased understanding that a very precise prediction model does not necessarily yield the greatest business value. As organizations are increasingly relying on data-driven decision-making, choosing the right evaluation metrics has become as critical as designing accurate prediction algorithms.

5.1. Traditional Classification Evaluation Metrics

Most of the customer churn prediction studies conducted earlier focused on traditional classification metrics from the confusion matrix to assess machine learning models. These metrics, such as Accuracy, Precision, Recall, F1-score and Area Under the Receiver Operating Characteristic Curve (ROC-AUC), gave a standardized way to compare the different classification algorithms. These evaluation metrics are suitable to measure the performance of a classification model on imbalanced datasets, as used by Sikri et al. Sikri et al. (2024) to benchmark various machine learning models for predicting customer churn in a telecom company.

Of these evaluation measures, Accuracy is one of the most commonly reported measures due to its simplicity and intuitive interpretation. It is the percentage of all of the instances that are correctly classified, and is therefore widely used in customer churn prediction research in the banking, telecom, subscription, and e-commerce sectors. But, there have been a number of researchers who have emphasized that in highly imbalanced customer churn data, Accuracy alone is not a complete measure of model performance. However, non-churning customers are typically the vast majority of observations and therefore a classifier can successfully classify a large number of customers without capturing many actual customers that churn Japkowicz and Stephen (2002), He and Garcia (2009), Branco, Torgo and Ribeiro (2016). As a result, this would cause an incorrect assessment of the performance of churn prediction models if this was the only metric used.

The problem with this is being overcome by researchers who started reporting Precision and Recall in addition to Accuracy. Precision is the percentage of churners the model predicts that are truly churners and Recall is the percentage of churners the model predicts that are actual churners. Increasing Recall tend to decrease Precision and vice versa, thus these two metrics are complementary. As a result, several studies have used the F1-score which is the harmonic mean of Precision and Recall to obtain a more complete picture of predictive performance. Sikri et al. Sikri et al.

(2024) have compared different Customer Churn Prediction models using the evaluation metrics – Accuracy, Precision, Recall, F1-score, ROC-AUC, and concluded that more than one evaluation metric is needed to measure the performance of the classifiers.

The Receiver Operating Characteristic (ROC) curve with respect to the area under the ROC curve (AUC) is another common metric used for evaluating a classification model. The ROC-AUC is a metric that is independent of the threshold value, unlike other metrics like Accuracy and F1-score, which are threshold dependent, and is especially useful when comparing different predictive models. Fawcett (2006) introduced ROC analysis as one of the recommended evaluation measures for a binary classification problem and Hand and Anagnostopoulos (2013) expanded the discussion on the cases where ROC-AUC gives a meaningful assessment of performance. Because of its threshold-independency, ROC-AUC is one of the most common metrics in the customer churn prediction literature, particularly when comparing multiple machine learning and deep learning models.

These traditional evaluation metrics are important for assessing the classification performance but have some business constraints. Most importantly, they assume that all the prediction errors are equally important irrespective of the importance of customer profitability and retention cost. In reality, however, the mistake of misclassifying a high-value customer in practical customer relationship management can cost much more than the misclassification of a low-value customer. So, even two models with the same accuracy or ROC-AUC could yield vastly different business results upon deployment. This observation has inspired researchers to examine evaluation methods which also consider the class imbalance, and recently, business profitability, not just predictive accuracy.

5.2. Evaluation under Class Imbalance

The key issue in the prediction of customer churn is that the datasets in the real world are highly imbalanced, with a large number of non-churning customers compared to churners. This imbalance can lead to inaccurate over-optimistic machine learning classifiers with very high Accuracy, but not learning anything about the true customers that will leave. Burez and Van den Poel (2009) were the first to draw attention to the fact that extremely imbalanced classes significantly reduce the efficacy of churn prediction models and stressed that detecting the minority classes is more important to the overall effectiveness of the churn prediction model than maximizing classification accuracy. Later, He and Garcia (2009) systematically studied the effect of imbalanced data on machine learning algorithms and showed that traditional classifiers are usually not very sensitive to the minority classes when the data is imbalanced without data preprocessing and/or algorithm adjustment.

Much research has been devoted to pre-training data-level balancing methods to address this problem. While the

minority samples are simply replicated, SMOTE takes advantage of the synthesis of existing minority-class samples to prevent overfitting and help balance the classes. Han *et al.* Han, Wang and Mao (2005) invented Borderline-SMOTE based on SMOTE, which can specifically generate synthetic samples near the decision boundary where misclassification is more likely to occur. He *et al.* He, Bai, Garcia and Li (2008) also proposed an adaptive Synthetic Sampling (ADASYN) algorithm which adaptively generates more synthetic samples for difficult-to-learn minority instances. Such methods are generally adopted as a part of customer churn prediction pipelines and have the advantage of being able to improve the detection of a minority class without modifying the underlying learning algorithms.

As mentioned above, the oversampling methods increase the balance between classes, but there are a number of researchers who have noted the shortcomings of these. However, synthetic oversampling could create noise or overlap in the data, especially when the data has a complex feature distribution, as pointed out by Branco *et al.* Branco *et al.* (2016). Likewise, Fernández *et al.* Fernández, García, Galar, Prati, Krawczyk and Herrera (2018) claimed that there is no resampling technique that dominates other resampling techniques for all application areas and candidates, and that the effectiveness of balancing techniques much relies on the datasets and classifiers. Amin *et al.* Amin, Anwar, Adnan, Nawaz, Howard, Qadir, Hawalah and Hussain (2016) also evaluated several oversampling methods for customer churn prediction, and found that the selection of oversampling strategy significantly impacts the performance of classifiers on imbalanced customer churn datasets. Thus, recent research has begun to integrate resampling, feature selection and cost-sensitive optimization into ensemble learning for boosting predictive robustness.

In addition to the balancing of the training data, researchers have also raised doubts about the applicability of conventional evaluation metrics in severe class imbalance. Saito and Rehmsmeier (2015) showed that Precision–Recall (PR) curves are a way to more accurately assess the prediction quality of the minority class in highly imbalanced binary classification problems. Their results have had significant impact on subsequent studies that predicted customer churn and where PR-AUC is now frequently reported with ROC-AUC, especially for datasets from telecom and subscription services that have relatively low churn rates.

The Matthews Correlation Coefficient (MCC) has been a heavily discussed evaluation metric. While both Accuracy and F1 scores take all four components of the confusion matrix into consideration, MCC compares the four components in a balanced manner, which makes it ideal when the class distributions are skewed. Chicco and Jurman (2020) showed that MCC can be a more accurate measure than Accuracy and F1-score for binary classification, and suggested that it is a key metric for evaluating imbalanced data. They found that their conclusions have

motivated researchers in customer churn prediction to consider MCC along with Precision, Recall and ROC-AUC in comparing predictive models.

Aside from resampling strategies, there have been algorithm-level ones gaining in popularity. These methods do not make changes to the training data—they simply make changes to how the learning is done, in the sense that they increase the punishment or cost for churners that are wrongly classified. Cost-sensitive learning, weighted loss functions, focal loss, and adaptive ensemble methods allow predictive models to focus on the minority class of customers and yet be competitive. In addition, the incorporation of misclassification costs directly into the learning objective greatly improves classification performance for the imbalanced data sets as shown by Sun *et al.* Sun, Wong and Kamel (2009) and more recent research has combined cost-weighted optimization with boosting and ensemble learning algorithms to further boost the performance of churn prediction.

However, despite all these developments, the evaluation of imbalances remains based on the premise that every churning customer is equal to another churning customer as far as the value for the organization is concerned. However, in the real world of customer relationship management, customers have varying buying habits, subscription values and profitability potential. The result is that a correct classification of a low value customer can have a much smaller impact on the organisation's performance than that of a high value customer. This rule has led to more research that takes imbalance-aware learning deeper into the business-focused evaluation area. In this area the main measures used to check how good a model is include customer lifetime value, the cost of campaigns and the expected money a business will make. These are more important, than just how accurate the model's.

5.3. From Statistical Accuracy to Business-Oriented Evaluation

Customer churn prediction models have always been evaluated using methods like Accuracy, F1-score and ROC-AUC to see how well they work.. Now researchers think that these methods may not be the best way to measure how well models perform in real business situations. For companies that manage customer relationships making much money as possible from their retention efforts is very important. They do not have a lot of money to spend on marketing and different customers are worth amounts. So the goal of predicting customer churn is no longer about being accurate. It is about making the money possible from customer retention.

The study by Verbraken *et al.* Verbraken *et al.* (2013) is one example of this way of thinking. They found that the old ways of measuring how good a model is do not show how money a company can make from keeping customers. To fix this they created the Expected Maximum Profit Measure for Customer Churn. This measure looks at how likely customersre to stay how much a campaign costs how much it takes to keep a customer and how much money a customer brings in. Their research showed that models that are very

accurate do not always make the money. This led to a way of evaluating models by looking at profit instead of just numbers.

Another group, led by Höppner *et al.* Höppner *et al.* (2020) created something called Profit-driven Decision Trees. This is a kind of decision tree that tries to make money directly while it is being built. Of using old methods like entropy or Gini index they used business goals to build the model. Their tests showed that models that used business goals made money than models that only looked at numbers. This shows that putting business goals into the model building process can help companies keep customers better and make money from customer retention. Customer churn prediction models like these can really help companies make the money possible, from their retention efforts.

Other industries also started to see the importance of making money when they build models. For example Maldonado *et al.* Maldonado, Domínguez, Olaya and Verbeke (2021) looked at customer churn in the mutual fund industry. They used customer data. Looked at how much each customer is worth. They didn't treat all customers the same. They focused on customers that're more valuable. This helped financial companies spend their money better. Their results showed that how much a customer is worth should be a part of how we check if a model is good.

More recent studies have taken this further. Gómez-Vargas *et al.* Gómez-Vargas *et al.* (2025) made a system that combines predicting if a customer will leave and figuring out the way to keep them. Of just looking at the chance a customer will leave they also looked at how much money the campaign will cost and how much money the customer is worth. This system helped make money than old ways of doing things.

Another group led by Feng *et al.* Feng, Yin, Wang, Ignatius, Cheng, Marra and Guo (2024) used a way of making decisions that helps with both predicting and making money. Their study showed that just picking customers based on their chance to leave doesn't always make the money. Some customers buy more. Bring in more money than others. By thinking about money while they made decisions they made money and still did a good job predicting.

Another important idea came introduced by Jiang *et al.* Jiang *et al.* (2024). They made a way to build models that help make money and also make sense. Unlike models that treat all customers the same their method gave more weight to customers that are more valuable. This helped models make decisions and also be easier to understand. Their tests showed they made money than old models and were still clear.

All these studies show that looking at money has grown beyond new ways to measure models. Now models use prediction making decisions knowing how much a customer is worth and helping make choices together. Customer churn is not about guessing who will leave it's about making smart choices that help the company make more money over time.

With these improvements most ways of making money still don't think about fairness for all customers. Some customers might not get the attention if the focus is only on money. So research is now looking at ways to make money and fairness are both considered. This new idea is covered next by looking at how much a customer's worth and how to check profit, for each customer.

5.4. Customer Lifetime Value and Individualized Profit Evaluation

As customer churn prediction has evolved from a purely predictive task to a business decision-support problem, Customer Lifetime Value (CLV) has become one of the most influential factors in evaluating customer retention strategies. Rust et al. Rust, Lemon and Zeithaml (2004) established customer equity as a central concept for aligning marketing strategy with long-term customer value, while Fader et al. Fader, Hardie and Lee (2005) proposed probabilistic models for estimating individual customer purchasing behaviour as an alternative to traditional CLV estimation. Rather than treating all customers equally, CLV estimates the long-term economic value that each customer is expected to generate throughout their relationship with an organization.

Early profit-driven frameworks generally relied on average customer revenue or population-level assumptions when estimating the financial impact of customer churn. Although these approaches demonstrated the importance of integrating business objectives into predictive modeling, they were unable to capture the substantial variation in profitability among individual customers. As a result, retention decisions derived from these models often overlooked valuable customers whose future contribution differed significantly from the population average Verbraken et al. (2013).

To address this limitation, several researchers began integrating individualized customer valuation into churn prediction models. Maldonado et al. Maldonado et al. (2021) developed a profit-driven framework for the mutual fund industry that combined churn prediction with customer segmentation and profitability analysis. Their study demonstrated that considering customer value during retention planning produced considerably higher financial returns than targeting customers solely according to predicted churn probability. In the way Jiang et al. Jiang et al. (2024) published a profit-focused weighted classifier. In this approach customer value played a role in the learning process by using adaptive sample weighting. When compared with classifiers their approach led to better business results while keeping the prediction process clear and easy to understand.

Recent studies have expanded on customer churn prediction that takes CLV into account by adding optimization methods to decision making. Gómez-Vargas et al. Gómez-Vargas et al. (2025) created a and-optimize framework. This framework takes into account churn probability, customer profitability, campaign limits and marketing budgets. Of making a churn prediction first and then choosing customers for retention their framework combines both steps to increase overall business profit. In a way Feng et al. Feng

et al. (2024) added prescriptive analytics to e-commerce customer retention. They showed that combining models with optimization methods greatly improves campaign success compared to older methods that only used probability-based customer rankings.

Another big step forward is in estimating customer value that changes over time. Traditional CLV methods usually assume that customer relationships stay the same and use sales as a sign of future value.. Customer behavior keeps changing because of new preferences, service quality and competition. Because of this recent studies have looked at analysis to estimate customer lifetime value. Kaplan-Meier estimators and other survival-based methods offer accurate views of future value by looking at the chance that customers stay active over time. These methods are becoming more popular because they handle data naturally and give estimates that change over time.

Based on these advances Manzoor et al. Manzoor et al. (2026) introduced the eProfits evaluation framework. This is one of the complete profit-focused evaluation methods ever made. Unlike the Expected Maximum Profit Measure, which uses assumptions about the group to estimate campaign profit e-Profits looks at profit on a customer level. It brings together survival-based CLV, individual retention costs, costs of action and chances of churn into one system. Their tests showed that models ranked with e-Profits were often different from those ranked with methods like Accuracy, F1-score and ROC-AUC. In some test data models that were not as good at predicting made more money because they focused more on customers who brought in real money.

More and more people are using profit evaluation. This shows a change in customer analytics. It is moving from predicting what will happen to giving advice on what to do. Of just finding customers who might leave new systems try to decide which customers should stay how to use limited marketing money and which plan will give the most value over time. So recent research is mixing churn prediction, with optimization tools, business systems, explainable AI and automated machine learning. The goal is to offer advice instead of just chances of events. Despite these advances, important research challenges remain. Accurate CLV estimation requires sufficient historical behavioural data, while many profit-driven evaluation frameworks depend on assumptions regarding campaign cost, retention success rate, and future purchasing behaviour that may vary considerably across organizations. Furthermore, optimization strategies that prioritize highly profitable customers may unintentionally disadvantage customers with lower historical spending or minority customer groups. Such bias raises important ethical concerns regarding equitable customer treatment and highlights the need to balance profitability with fairness in intelligent customer retention systems. Therefore, Recent studies indicate that customer churn prediction has evolved from accuracy-oriented classification toward customer-level profit optimization. Table 4 summarizes the major profit-driven evaluation frameworks proposed in the

Table 4

Comparison of representative profit-driven customer churn evaluation frameworks.

Year	Reference	Method / Framework	Main Contribution	Limitation
2026	Manzoor et al. (2026)	e-Profits	Introduced a customer-level profit evaluation framework by integrating survival-based CLV, individualized retention costs, and expected financial return.	Requires reliable customer-level historical data and survival analysis for accurate CLV estimation.
2025	Gómez-Vargas et al. (2025)	Predict-and-Optimize	Jointly optimized churn prediction and retention campaign decisions to maximize overall business profit under operational constraints.	Computational complexity increases with large-scale optimization problems.
2024	Feng et al. (2024)	Prescriptive Analytics	Combined predictive analytics with campaign optimization to improve profitability in e-commerce customer retention.	Performance depends on business assumptions and campaign parameter settings.
2024	Jiang et al. (2024)	Profit-Driven Weighted Classifier	Assigned adaptive weights according to customer value, improving both profitability and model interpretability.	Requires reliable customer valuation before model training.
2021	Maldonado et al. (2021)	CLV-based Churn Prediction	Integrated customer segmentation and profitability analysis into churn prediction for targeted retention.	Validated only in the mutual fund industry.
2020	Höppner et al. (2020)	ProfTree	Optimized decision tree construction directly using expected business profit instead of classification accuracy.	Limited to tree-based learning algorithms.
2013	Verbraken et al. (2013)	EMPC	Proposed the Expected Maximum Profit Measure (EMPC), demonstrating that business profit is a more meaningful evaluation criterion than traditional classification metrics.	Relies on population-level assumptions regarding customer behavior and campaign success.

literature, highlighting their objectives, strengths, and remaining limitations. Although the frameworks summarized in Table 4 significantly improve the financial effectiveness of customer retention campaigns, nearly all of them optimize business profit as the primary objective. Relatively little attention has been given to whether these profit-maximizing strategies treat different customer groups fairly or inadvertently introduce algorithmic bias. As customer analytics increasingly influences business decision-making, balancing profitability with fairness has become an important research direction. The following section reviews fairness-aware customer churn prediction methods and discusses recent approaches for integrating ethical AI principles into profit-driven retention systems.

6. Trustworthy Artificial Intelligence for Customer Churn Prediction

Machine learning and deep learning models are really good at figuring out when customers are going to leave.. These models are getting more and more complicated. This

is causing some problems. We need to make sure these models are transparent, fair and that we can hold them accountable.

Models like Random Forest, XGBoost LightGBM and deep neural networks are very good at predicting when customers will leave. However it is hard for the people making business decisions to understand why these models are making predictions. These models are like boxes that we cannot see inside of. They just give us answers without explaining how they got those answers. Machine learning and deep learning models need to be more straightforward so business decision-makers can understand the reasons behind the predictions, about customer churn.

In practical customer retention environments, prediction accuracy alone is insufficient. Organizations require churn prediction systems that can explain decisions, avoid discriminatory outcomes, protect customer interests, and support trustworthy decision-making. Therefore, Trustworthy Artificial Intelligence (Trustworthy AI) has emerged as an important research direction that combines explainability, fairness, robustness, privacy, and accountability.

A trustworthy churn prediction framework can be represented as:

$$\begin{aligned} \text{Trustworthy AI} = & \\ & \text{Accuracy} \\ & + \text{Explainability} \\ & + \text{Fairness} \\ & + \text{Accountability} \end{aligned} \quad (9)$$

6.1. Explainable Artificial Intelligence for Customer Churn Prediction

Explainable Artificial Intelligence (XAI) aims to provide understandable explanations for machine learning predictions while maintaining predictive performance. Guidotti et al. (2019) described XAI as a framework for interpreting black-box machine learning models and improving human understanding of automated decisions.

In customer churn prediction, explainability enables organizations to identify the factors responsible for customer departure risk. Important churn-related factors may include:

- Customer tenure,
- Contract duration,
- Monthly charges,
- Service complaints,
- Payment behaviour,
- Usage patterns.

Arrieta et al. (2020) emphasized that explainability is a fundamental requirement for responsible AI because high-performing models without interpretation may create trust and adoption barriers.

Similarly, Adadi and Berrada (2018) reviewed major XAI approaches and highlighted that transparency is particularly important in high-impact decision-making applications.

Recent research has further focused on evaluating explanation quality rather than only generating explanations. Doshi-Velez and Kim (2017) proposed evaluation principles for interpretable machine learning systems, emphasizing human understanding, explanation completeness, and application-based evaluation.

6.2. SHAP and LIME-Based Explanation Methods

When we look at the ways to explain machine learning predictions, SHAP, which stands for SHapley Additive exPlanations is one of the most popular methods.

SHAP is used to explain machine learning predictions. Lundberg and Lee (2017) introduced SHAP, which is based on Shapley values from a part of mathematics called cooperative game theory. This method helps us understand how each feature affects the prediction outcome.

The way SHAP explains things can be shown with an equation:

$$f(x) = \phi_0 + \sum_{i=1}^M \phi_i \quad (10)$$

where ϕ_i represents the contribution of feature i toward the prediction.

Lundberg et al. (2020) demonstrated that SHAP is good at providing explanations for certain types of models like tree-based models, which are often used in systems that predict customer churn, such as XGBoost, Random Forest and LightGBM.

There is another method that people use a lot called Local Interpretable Model-Agnostic Explanations or LIME for short. Ribeiro et al. (2016) introduced LIME as a way to explain predictions by using simpler models that are easier to understand.

Molnar (2022) summarized several model-agnostic explainability techniques, including feature importance analysis, partial dependence plots, local surrogate models, and SHAP, highlighting their importance for interpreting complex machine learning models in practical applications.

6.3. Bias and Fairness Challenges in Customer Churn Prediction

Although AI-based churn models improve customer analytics, they may unintentionally learn biases present in historical customer data. Since churn prediction models are trained using demographic information, financial characteristics, service usage patterns, and customer interactions, biased historical decisions may influence future predictions.

Mehrabi et al. (2021) identified several sources of algorithmic bias, including biased datasets, biased feature selection, biased model assumptions, and unequal prediction outcomes.

In customer churn prediction, potential sources of bias include:

- Unequal representation of customer groups,
- Historical bias from previous business decisions,
- Proxy variables representing sensitive attributes,
- Biased churn labels,
- Feedback loops caused by automated decisions.

Barocas et al. (2019) explained that machine learning systems may reproduce existing organizational biases when trained using historical decision data.

Suresh and Gutttag (2021) proposed a lifecycle-based framework showing that bias can occur during data collection, feature engineering, model development, evaluation, and deployment.

6.4. Fairness Metrics for Churn Prediction

Fairness evaluation determines whether churn prediction models provide equitable outcomes among different customer groups.

The most commonly used fairness criteria include:

6.4.1. Demographic Parity

Demographic parity requires prediction outcomes to be independent of sensitive attributes:

$$P(\hat{Y} = 1|A = a) = P(\hat{Y} = 1|A = b) \quad (11)$$

6.4.2. Equal Opportunity

Equal opportunity requires similar true positive rates across different groups:

$$TPR_a = TPR_b \quad (12)$$

Hardt et al. Hardt, Price and Srebro (2016) introduced equality of opportunity as a fairness criterion for supervised learning systems.

6.4.3. Equalized Odds

Equalized odds considers both true positive and false positive rates:

$$TPR_a = TPR_b \quad (13)$$

and

$$FPR_a = FPR_b \quad (14)$$

Pessach and Shmueli Pessach and Shmueli (2022) reviewed fairness-aware machine learning approaches and highlighted the challenges of balancing fairness and predictive performance.

6.5. Fairness-Aware Learning and Bias Mitigation

Fairness-aware learning approaches can be categorized into three major groups:

6.5.1. Pre-processing Approaches

Pre-processing techniques modify datasets before model training through methods such as re-sampling, re-weighting, and bias correction.

6.5.2. In-processing Approaches

In-processing approaches incorporate fairness constraints directly into the learning objective:

$$Loss = PredictionLoss + \lambda FairnessPenalty \quad (15)$$

where λ controls the importance of fairness during optimization.

Subgroup-fairness methods can be used to audit for unfair outcomes and also train classifiers that adhere to fairness across structured subgroups Kearns, Neel, Roth and Wu (2018).

6.5.3. Post-processing Approaches

Post-processing approaches change the results a model gives after it has been trained in an effort to make predictions more equal for everyone.

Zafar et al. Zafar, Valera, Rodriguez and Gummadi (2017) created fairness rules while the model was being built showing that fairness goals can be added directly to machine learning systems.

Bellamy et al. Bellamy, Dey, Hind, Hoffman, Houde, Kannan et al. (2019) made the AI Fairness 360 toolkit, which offers real-world tools for finding and fixing unfairness in computer algorithms.

6.6. Balancing Profitability, Explainability, and Fairness

A big problem when trying to predict customer churn for money-making reasons is finding the balance between what's good for the business and what's fair. Models that make the most money might focus on customers who bring in the cash but systems that care about fairness try to treat all customer groups equally.

So the way to handle this in the future is to work on making sure the models do well in predicting churn make money are easy to understand and're fair to everyone.

Objective =

$$\begin{aligned} & \text{Accuracy} + \text{Profitability} \\ & + \text{Explainability} + \text{Fairness} \end{aligned} \quad (16)$$

Using AI to predict churn in a way is a big change. Of just looking at customer risk it's, about creating systems that help make decisions in a way that is clear, fair and matches what the company wants.

7. Proactive Customer Retention and Prescriptive Analytics

Customer churn prediction is usually about finding out which customers will stop using a service.. Just knowing if a customer is likely to leave does not tell us what to do to keep them. So now people are working on systems that not find out who might leave but also suggest what actions to take to keep them.

These systems use intelligence to find customers who might stop using a service and to recommend the best way to keep them. According to Verbeke et al. Verbeke et al. (2011) managing customer churn is not about being correct when we say who will leave. We need to think about what each customer's, like what we can do to keep them and what we hope to get out of it. This way of thinking has encouraged people to make systems that combine predicting who will leave with finding the best way to keep them.

7.1. From Predictive Analytics to Prescriptive Retention

Predictive analytics looks at how likely a customer's to leave while prescriptive analytics figures out what the best

step is to stop them from leaving. Modern systems that keep customers come up with plans that take into account how likely a customer is to leave how much that customer is worth over time how much it will cost to try to keep them. How likely they are to respond to an effort to keep them. All of these factors help choose the possible strategy, for the business.

Burez and Van den Poel Burez and Van den Poel (2009) showed that how well a company manages customers leaving depends not on finding the customers who are most likely to leave but also on choosing which customers to target for retention activities based on how profitable those activities will be.

So the goal of retention systems can be written like this:

$$Optimal\ Action = \operatorname{argmax}(Expected\ Profit(Action)) \quad (17)$$

This means the best action is the one that gives the expected profit. The best way to keep a customer is the one that brings the economic value.

7.2. Reinforcement Learning for Retention Optimization

Reinforcement Learning has recently become popular as a way to help keep customers because it lets models figure out the actions by talking and learning from customers over time.

In contrast to methods that only guess if a customer will leave Reinforcement Learning looks at how customers act what can be done to keep them and what happens in the long run.

The Reinforcement Learning setup includes:

- State: What the customer's like right now and how they behave
- Action: Things that can be done to keep the customer like giving a discount making service better or sending messages
- Reward: The money gained from keeping the customer.

The total reward that is expected can be shown as:

$$G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k} \quad (18)$$

where R_t represents the reward obtained at time t and γ represents the discount factor.

Recent work has found that Reinforcement Learning can help with making decisions over time when customer reactions change. Sutton and Barto Sutton and Barto (2018) provided the ideas of Reinforcement Learning and more recent studies have looked at using it for personalized suggestions and keeping customers.

7.3. Personalized Retention Strategies

Modern customer retention strategies focus more and more on personalization because each customer has reasons for leaving. AI systems look at customer details past purchases how they interacted with support and how they behave to create retention plans for each person.

Lemmens and Gupta Lemmens and Gupta (2020) said that using predictions about who might leave along with grouping customers makes retention efforts stronger. This helps companies focus on the customers who matter most.

Personalized retention efforts often include:

- Customer segmentation,
- Recommendation systems,
- Sentiment analysis,
- Predicting how much a customer is worth over time
- Custom marketing actions.

7.4. Real-Time Churn Analytics

More and more customer data is coming in time which has led to the creation of real-time churn prediction tools. Older models use data and get updated now and then which might not catch quick changes in how customers act in the market or in how they interact with services.

Real-time churn analytics solves this problem by always gathering, looking at and studying customer data to spot early signs that a customer might leave. These tools let companies switch from reacting to problems to taking action before things get worse.

New frameworks, for managing churn use AI, real-time analysis and automatic decisions to find signs of churn and suggest quick actions. Ip Ip (2025) showed that today's churn management is changing from models to new more flexible and custom plans.

Real-time churn analytics commonly utilizes:

- Online learning algorithms,
- Streaming data processing,
- Deep learning architectures,
- Customer interaction monitoring,
- Real-time behavioural feature extraction,
- Adaptive prediction models.

Dynamic customer environments create a major challenge because customer preferences and behavioural patterns continuously evolve. Gama et al. Gama, Zliobaite, Bifet, Pechenizkiy and Bouchachia (2014) highlighted that concept drift can reduce model performance when prediction systems are deployed in changing environments.

Recent studies have explored adaptive machine learning and deep learning approaches for continuously updating churn prediction models. Bifet et al. Bifet, Gavaldà, Holmes

and Pfahringer (2009) investigated adaptive learning methods for evolving data streams, demonstrating the importance of continuously updated models in dynamic applications.

Deep learning approaches have further improved real-time churn analytics by capturing sequential customer interactions. The long-term memory (LSTM) architecture was developed to capture long-range dependencies in sequences Hochreiter and Schmidhuber (1997). This offers a methodological basis for modelling sequential user behaviours.

Recent research has applied recurrent neural networks, attention mechanisms, and transformer-based architectures for modelling customer activity sequences and predicting future churn risk Wu et al. (2020).

Furthermore, large-scale real-time churn systems require distributed and privacy-preserving architectures. Federated learning approaches have been explored to enable collaborative analytics without centralized storage of sensitive customer information McMahan et al. (2017); Coimbra et al. (2024)

However, several challenges remain in real-time churn analytics:

- Rapid changes in customer preferences,
- Concept drift,
- Limited real-time labelled data,
- Privacy risks during continuous monitoring,
- Computational requirements for real-time inference.

Therefore, future real-time churn prediction systems should integrate adaptive learning, deep sequential models, explainable AI, privacy-preserving mechanisms, and business-oriented optimization to support intelligent customer retention.

7.5. Challenges of Proactive Retention Systems

Although proactive retention systems provide significant opportunities for improving customer relationship management, several important challenges remain.

• **Difficulty in Measuring the Causal Effect of Retention Actions:**

Traditional churn prediction models estimate the probability of customer departure but do not determine whether a specific retention action will actually change customer behaviour. Therefore, causal inference and uplift modelling approaches are required to estimate the true impact of retention interventions.

Gutierrez and Gérardy Gutierrez and Gérardy (2017) highlighted that uplift modelling provides a framework for identifying customers who are most likely to respond positively to specific interventions rather than simply predicting churn probability.

• **Limited Availability of Intervention-Response Datasets:**

A major challenge in proactive retention is the lack of high-quality datasets containing information about customer responses to different retention strategies. Most publicly available churn datasets only contain historical customer characteristics and churn labels without information about previous interventions.

As highlighted by Verbeke et al. Verbeke et al. (2011), effective churn management requires linking prediction outcomes with actionable retention strategies and customer responses.

• **Balancing Profitability and Customer Fairness:**

Although profit-driven retention systems aim to maximize business value, optimizing only financial outcomes may result in unequal treatment of customer groups. Therefore, proactive retention systems should consider fairness constraints together with profitability objectives.

Mehrabi et al. Mehrabi et al. (2021) emphasized that automated decision systems must evaluate potential bias and ensure fair treatment across different demographic groups.

• **Continuous Changes in Customer Preferences:**

Customer behaviour is dynamic and may change due to market conditions, competitor actions, economic factors, and evolving customer expectations. Therefore, proactive retention systems require adaptive learning mechanisms capable of handling changing data distributions.

Gama et al. Gama et al. (2014) discussed concept drift as a major challenge in machine learning systems operating in dynamic environments, where relationships between input variables and target outcomes continuously change.

• **Privacy Concerns During Personalization:**

Personalized retention strategies require extensive customer information, including behavioural, transactional, and interaction data. However, collecting and processing such sensitive information introduces privacy and security concerns.

Federated learning has emerged as a privacy-preserving alternative that enables collaborative model training without directly sharing raw customer data. Recent federated learning studies have addressed challenges such as distributed data heterogeneity, communication efficiency, and privacy preservation, while churn-related research has explored federated approaches for privacy-aware customer analytics McMahan et al. (2017); Coimbra et al. (2024).

Therefore, future churn prediction research should move beyond isolated prediction models toward adaptive and intelligent retention ecosystems that integrate predictive analytics, causal inference, explainable AI, fairness-aware

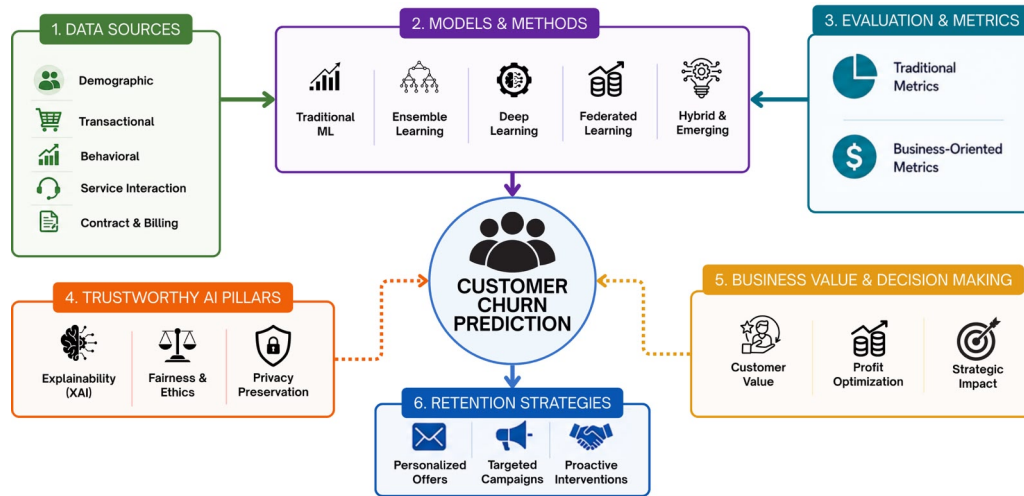


Figure 2: Unified Intelligent Churn Management Framework integrating predictive intelligence, business value optimization, trustworthy artificial intelligence, and proactive retention strategies.

decision-making, privacy preservation, and business optimization. Churn prediction systems can become useful tools that help companies make good decisions. They can do this by combining a few parts. This way companies can create strategies to keep customers that're fair and make money.

To explain how this works we made a plan called the Unified Intelligent Churn Management Framework. You can see what it looks like in Figure 2. This framework uses customer data gets the data ready and then uses predictions to help companies make good decisions. It also thinks about what's good for business uses artificial intelligence that people can trust and finds ways to keep customers.

The old way of doing things was about finding out who might leave.. Now we want to keep customers and make sure everyone is treated fairly. We also want to make sure companies make money and keep customers for a time.

Our plan shows how companies can use customer information to make decisions about keeping customers. It uses smart layers to turn customer information into actions that keep customers. First we use models to figure out who might leave. Then we use techniques to see how much each customer is worth and if they are profitable. We also use intelligence that is fair and transparent. Finally we use what we learned to create plans to keep customers. This helps companies keep customers make money from each customer and grow their business in a way that is good, for everyone. The Unified Intelligent Churn Management Framework is a way to manage churn and keep customers..

8. Cross-Domain Churn Analysis: Applied Features and Techniques

Customer churn prediction is something that a lot of people have looked into in different industries. This is because customers leaving a company can be affected by things that're specific to that industry what customers expect how competitive the market is and how good the service is. Even

though machine learning and deep learning can be used in different areas being able to predict customer churn really depends on having the right information about customers understanding the business and knowing what the company wants to achieve with customer retention.

Some recent studies have shown that it is unlikely that one customer churn prediction model will work best for all industries. This is because different industries have customers that behave in ways and make decisions differently. So looking at how different industries handle customer data, predictive models and retention strategies can provide a lot of information. This can help organizations improve the way they manage their relationships, with customers. Customer churn prediction is an area of study and customer churn prediction models need to be tailored to each industry. Customer churn prediction is an issue and companies need to understand customer churn prediction if they want to keep their customers.

8.1. Telecommunication Churn Prediction

Telecommunication remains one of the most extensively studied domains in customer churn prediction because high switching rates directly influence revenue stability, customer lifetime value, and market competitiveness. Due to intense competition among telecom operators, identifying potential churners before service termination has become a critical analytical objective.

Telecom churn prediction models commonly utilize behavioural, demographic, and service-related features, including:

- Customer tenure,
- Contract duration,
- Monthly charges,
- Call duration and internet usage,
- Service complaints,

- Billing information,
- Customer support interactions,
- Network quality indicators.

Early customer churn studies mainly focused on statistical approaches and traditional machine learning algorithms. Hadden et al. Hadden et al. (2007) reviewed computer-assisted churn management approaches and highlighted the role of data mining techniques in identifying customers at high risk of churn.

Similarly, Coussement and Van den Poel Coussement and Van den Poel (2008) applied Support Vector Machines to churn prediction in a subscription-service context and showed that subscription-related variables were important predictors, while customer–company interaction variables also contributed to churn prediction.

With the rapid advancement of machine learning, ensemble learning techniques have become increasingly popular for customer churn prediction because they combine the strengths of multiple base classifiers to improve predictive performance. Wahul et al. Wahul et al. (2023) evaluated several ensemble methods, including Random Forest, AdaBoost, Gradient Boosting, and Stacking, and demonstrated that the stacked ensemble achieved the highest Accuracy, Precision, Recall, F1-score, and ROC-AUC among the evaluated models. Their findings indicate that ensemble learning can provide more robust and accurate churn prediction than individual classifiers.

Sikri et al. Sikri et al. (2024) further demonstrated that effective feature engineering and comprehensive model evaluation can significantly improve the reliability of telecom customer churn prediction.

Recent studies have expanded telecom churn analysis through deep learning, explainable AI, and large-scale analytics. Barsotti et al. Barsotti et al. (2024) provided a recent survey of telecom churn prediction techniques and highlighted the evolution from traditional machine learning toward advanced artificial intelligence approaches.

Mettle et al. Mettle et al. (2026) investigated machine learning-based telecom customer churn prediction by comparing multiple classification algorithms, demonstrating the effectiveness of machine learning techniques for identifying potential churners and supporting customer retention strategies.

Explainability has become increasingly important in customer churn prediction because organizations require transparent explanations before applying retention interventions. Lundberg et al. Lundberg et al. (2020) demonstrated how SHAP-based methods can provide both local and global explanations for tree-based machine learning models. In the context of customer churn prediction, Maan and Maan Maan and Maan (2023) applied explainable machine learning and Shapley-value-based feature analysis to identify important factors influencing churn predictions and improve model transparency.

However, telecom churn prediction continues to face challenges including severe class imbalance, rapidly changing customer behaviour, privacy concerns, and model transferability across different markets. Recent studies have also explored composite deep learning architectures Khattak et al. (2023), comparative machine learning models specifically for telecom churn Nadeem, Ramachandran and Rajeswari (2021), explainable machine learning frameworks for telecom churn Yuan and Liu (2025), AI-driven predictive frameworks for enhanced customer relationship management Abdelhady and Mohamed (2025), and data-driven telecom marketing optimization combined with churn prediction and customer segmentation Ali, Ahmed and Gasmalla (2026). Future telecom churn systems should therefore integrate predictive accuracy, profitability optimization, explainability, fairness, and privacy-preserving learning approaches.

8.2. Banking and Financial Services Churn Prediction

Banking churn prediction has received significant research attention because customer retention directly influences profitability, customer lifetime value, and long-term financial relationships.

Unlike telecom services, banking churn is strongly associated with financial behaviour, account activity, service usage, and relationship characteristics.

Common predictive features include:

- Account balance,
- Transaction frequency,
- Credit history,
- Loan information,
- Digital banking activity,
- Customer satisfaction,
- Product ownership.

More recent studies have incorporated additional customer information into machine learning models to improve banking churn prediction. De Caigny et al. De Caigny et al. (2020) investigated the use of additional information in customer churn prediction and demonstrated its potential to improve predictive performance.

Explainable artificial intelligence has also become important in banking churn prediction because financial institutions require transparent and interpretable decision-making. Guidotti et al. Guidotti et al. (2019) reviewed different approaches for explaining black-box machine learning models and highlighted the importance of model interpretability. For tree-based models, Lundberg et al. Lundberg et al. (2020) demonstrated that SHAP-based explanations can provide both local and global interpretations of model predictions. These explanations can help us figure out what makes a customer leave and support decisions on how to keep customers.

Banking is a case when it comes to predicting if a customer will leave. Banking churn prediction has to deal with a lot of rules like what information can be shared and making sure the bank is being fair to its customers. So when we make banking churn models we need to make sure they are fair and good for business which means banking churn models should use banking churn principles that people can trust and also think about what is good, for the bank, which is banking churn.

8.3. Small and Medium Enterprise (SME) Churn Prediction

SME churn prediction presents unique challenges because smaller organizations usually possess limited customer records, fewer behavioural variables, and restricted analytical resources compared with large enterprises. Unlike large-scale industries such as telecommunications and banking, SMEs often operate with smaller customer populations, making customer retention decisions highly sensitive to individual customer relationships.

Ahmad and Islam [Ahmad and Islam \(2021\)](#) investigated customer churn prediction for small and medium enterprises (SMEs) using machine learning techniques and showed that machine learning models can be effectively used to identify potential customer churn.

SME churn prediction commonly considers customer-related and behavioural information such as:

- Customer transaction history,
- Relationship duration,
- Service usage patterns,
- Customer interaction frequency,
- Purchase behaviour,
- Contract information.

Customer relationship characteristics are particularly important in SME environments because losing existing customers can affect revenue stability. Reinartz and Kumar [Reinartz and Kumar \(2000\)](#) studied the relationship between customer relationship duration and profitability and showed that longer customer relationships do not always result in higher profitability.

Furthermore, ensemble learning approaches have been explored to improve prediction performance when customer datasets are limited. Recent customer analytics research suggests that combining churn probability with customer value estimation can improve retention prioritization.

However, SME churn modelling remains challenging because of:

- Limited historical customer data,
- Small sample size,
- Class imbalance,

- Lack of behavioural diversity,
- Limited computational resources.

Therefore, future SME churn prediction systems should focus on lightweight, interpretable, and business-aware artificial intelligence frameworks suitable for resource-constrained environments.

8.4. SaaS and OTT Platform Churn Prediction

Subscription-based digital platforms such as Software-as-a-Service (SaaS) and Over-The-Top (OTT) services experience continuous customer engagement changes because users can easily switch providers with minimal switching costs.

Unlike traditional industries, SaaS and OTT churn is strongly influenced by digital interaction patterns, content consumption behaviour, and subscription engagement.

Important predictive features include:

- Login frequency,
- Subscription duration,
- Content consumption patterns,
- Payment behaviour,
- User engagement metrics,
- Feature utilization.

Kumar and Gupta [Devi et al. \(2023\)](#) demonstrated that deep learning approaches can capture complex behavioural patterns in SaaS and OTT churn prediction.

Vafeiadis et al. [Vafeiadis et al. \(2015\)](#) compared several machine learning techniques for customer churn prediction in the telecommunications industry. Their results showed that boosted versions of the classifiers generally outperformed their non-boosted counterparts, demonstrating the effectiveness of ensemble-based approaches for identifying customers at risk of churn.

Transformer-based architectures have recently gained attention because of their ability to capture long-term dependencies in sequential customer behaviour.

However, SaaS and OTT churn prediction remains challenging due to rapidly changing preferences, evolving content consumption patterns, and limited interpretability of complex deep learning models.

8.5. E-Commerce Customer Churn Prediction

E-commerce churn prediction differs from traditional service industries because customer decisions are influenced by purchasing behaviour, product satisfaction, digital interactions, and recommendation experiences.

Common predictive features include:

- Purchase frequency,
- Average order value,

- Product reviews,
- Browsing behaviour,
- Recommendation interactions,
- Customer engagement history.

Customer analytics research has demonstrated the importance of considering customer value when making retention decisions. Gupta et al. (2004) discussed customer lifetime value (CLV) as an important measure for evaluating the long-term economic value of customers and supporting customer-focused marketing and retention strategies.

Recent studies have also explored the use of data mining techniques in customer relationship management. Ngai et al. (2009) reviewed data mining applications in CRM and highlighted their role in analysing customer information, identifying customer patterns, and supporting customer relationship and retention decisions.

Recent e-commerce churn studies have demonstrated that transaction behaviour, customer engagement patterns, and purchasing history are important predictors of customer retention outcomes. Machine learning-based customer analytics approaches have been widely applied to identify behavioural patterns associated with customer disengagement.

However, e-commerce churn prediction faces challenges including rapidly changing customer preferences, high-dimensional behavioural data, and the difficulty of distinguishing temporary inactivity from permanent customer loss. De Alwis and Ekanayake (2025) addressed some of these challenges by integrating explainability, risk modelling, and customer segmentation into a unified churn analytics framework for personalized retention in e-commerce settings.

8.6. Healthcare Churn Prediction

Healthcare organizations are using churn prediction methods more and more to understand why patients stop coming improve how they keep patients coming back and make plans to keep patients around.

In areas churn prediction is not the same as in healthcare. Healthcare churn prediction has complicated parts like medical details actions patients take and things related to services because patients might decide to stop based on how they feel about treatment how easy it is to get care how good the care is and how happy they are.

Healthcare models often look at:

- Patient visit frequency,
- Treatment history,
- Healthcare utilization patterns,
- Appointment behaviour,
- Patient satisfaction,
- Insurance information.

Zhao et al. (2021) made machine learning models to predict when patients might come back to the hospital early showing that smart data can find people at risk and help doctors act faster.

Rajkomar et al. (2019) showed how machine learning and deep learning can find patterns in big healthcare data. Beam and Kohane (2018) talked about how big data and machine learning're changing how doctors make choices pointing out the good and the bad of using smart data with private patient details. These smart tools can help find patients who need help.

Explainability is also important in healthcare because the decisions doctors make need to be clear and easy to understand. Tjoa and Guan (2021) looked at how to make AI easier to understand in healthcare. Said that being able to explain AI is important for making smart choices.

Also keeping things is important because healthcare data has personal details. Rieke et al. (2020) talked about how using a method called learning can help build models without putting all the data in one place, which helps keep things safe.

Overall research in healthcare analytics shows that combining models being able to explain them and keeping data safe is important when making smart systems for patient decisions.

Healthcare churn prediction is still hard because there are not enough labeled data, privacy rules, different types of data and ethical issues.

Insurance companies are using churn prediction models more to find customers who might stop their policies switch to companies or get less involved. Keeping customers is important for making money so using tools to predict and keep customers is a big part of how insurance companies manage their customers.

Important things to look at include:

- Policy duration,
- Claim history,
- Premium information,
- Customer demographics,
- Payment behaviour,
- Customer service interactions,
- Policy renewal history.

Verbeke et al. (2011) said that models that are easy to understand are important for managing customer relationships.

To explain the predictions of a model and the contributions of features, there are tools that can be used by the user. There are tools to help the user understand the model predictions and feature contributions, which are known as explainability methods Molnar (2022). These approaches

Table 5
Cross-Domain Applications of Customer Churn Prediction

Year	Domain	Important Features	Common Techniques	Challenges	Reference
2026	Telecommunication	Customer demographics, service usage, and account information	Machine learning classification	Class imbalance and model selection	Mettle et al. (2026)
2026	Telecommunications	IBM Telco and Maven Telecom datasets	Machine Learning and AI analytics	Generalization across industries	Manzoor et al. (2026)
2024	Telecommunication	Customer demographics, service usage, account information	Machine learning classification and predictive modelling	Class imbalance and model performance evaluation	Sikri et al. (2024)
2024	Telecommunication	Large-scale customer interactions, behavioural signals	Deep learning and scalable analytics	Real-time processing	Barsotti et al. (2024)
2021	Healthcare (Hospital Readmission)	Demographics, laboratory tests, vital signs, diagnoses, medications, healthcare utilization	Logistic Regression, Random Forest, XGBoost, AdaBoost, Decision Tree, Neural Network	High-dimensional EHR data, class imbalance, limited generalizability across healthcare systems	Zhao et al. (2021)
2021	SME Services	Transaction history, relationship duration, service usage	Traditional ML models	Limited customer data availability	Ahmad and Islam (2021)
2021	Banking	Customer behaviour, interaction information, textual customer data	Machine Learning models	Data complexity and interpretability	De Caigny et al. (2020)
2009	Customer Analytics / E-commerce	Customer relationship management datasets	Customer behaviour, transaction information, interaction history	Data mining techniques and CRM analytics	Ngai et al. (2009)

could be applied to the problem of insurance churn prediction to aid in clear decisions on insurance retention.

These changes show that tools and being able to explain them are more important than ever in insurance churn prediction helping companies make better and clearer choices to keep their customers.

Insurance churn prediction has problems like different types of customers, privacy rules, rules that must be followed and needing to balance making money with treating customers fairly.

When looking at how churn prediction works in areas the research shows that what people do what features are important how models are built and the problems faced are different in each area. Table 5 shows some of the studies from areas between 2016 and 2025 showing what each industry is about the tools used and the main problems.

Looking across different areas shows that how churn prediction works depends a lot on the industry. Telecom and software companies use things like how customers act and how involved they're. Banking and insurance focus on money stuff. How much customers are worth. Healthcare has to think about keeping data safe and being fair because the data's very private. So the future of churn prediction should be, about systems that use what people know about each area make money explain things be fair and keep data safe.

9. Comparative Analysis of Customer Churn Prediction Studies

Customer churn prediction research has moved from ways of looking at data to more advanced methods that use artificial intelligence. These new ways include machine

learning, deep learning and other techniques that help explain why customers leave. Also there is focus on making sure these models are fair protect customer privacy and help businesses make more money. A recent study by Chaudhary et al. Chaudhary, Sharma and Pal (2025) shows this change using AI and big data to make predictions about customers leaving.. Different studies still have different areas of focus different types of data different ways to prepare the data, different models and different ways to check how good the models are.

To see how customer churn prediction has changed this section looks at some studies. Table 6 shows some studies from years. It includes the year, the type of business the data used the features looked at the methods used and the problems found. This table shows that the focus has changed from making better predictions to making better management systems that include things like explaining how the models work making sure they are fair and protecting customer data.

The comparison in Table 6 shows some changes in how researchers look at customer churn. Earlier studies mainly used machine learning to make predictions, especially in telecom and banking. More recent studies have used learning, explainable AI and methods that focus on profit and privacy.

Even though there have been changes in the methods used some problems still exist. Many studies still look at how accurate the models are but they don't pay enough attention to how profitable the models are, how fair they are, how easy they are to understand or how well they work in real situations. These problems show that there needs to be systems that combine good predictions with systems that are fair trustworthy and good, for business.

10. Research Challenges and Future Directions

Customer churn prediction has gotten a lot better with machine learning and other methods.. There are still some problems that need to be fixed. Some studies can predict when customers will leave well. However to actually use these systems in life we need to deal with some issues.

Customer churn prediction systems have to be good for business they have to be fair. They have to protect peoples privacy. We also need to be able to understand how they make decisions and use that information to make choices.

This part talks about the problems, with customer churn prediction that we found in other studies. It also looks at where we should go from here to make customer retention better. We want customer retention to be smart and responsible. We want customer churn prediction to be adaptive and intelligent. We want to make customer retention ecosystems that really work.

10.1. Limitations of Existing Churn Prediction Studies

10.1.1. Overemphasis on Predictive Accuracy

A major limitation of existing churn prediction research is the continued emphasis on classification-based evaluation metrics, including Accuracy, Precision, Recall, F1-score, and ROC-AUC. Although these metrics provide valuable information about predictive performance, they do not directly measure business effectiveness, customer value, intervention cost, or retention profitability.

Many studies focus on identifying customers with a high probability of churn without considering whether retention interventions will generate positive financial outcomes. Hoppner et al. Höppner et al. (2020) demonstrated the importance of incorporating economic considerations into churn management, while Verbraken et al. Verbraken et al. (2013) investigated profit-driven evaluation and optimization of customer churn prediction. More recently, Manzoor et al. Manzoor et al. (2026) proposed the e-Profits framework to evaluate churn prediction from a business-oriented perspective.

Therefore, future churn prediction systems should integrate customer lifetime value, profit optimization, and decision-oriented evaluation frameworks to better align predictive models with organizational objectives.

10.1.2. Static Modelling and Behavioural Changes

Customer behaviour continuously evolves due to changing market conditions, competitor strategies, pricing changes, and customer expectations. However, many existing churn prediction studies rely on static historical datasets and assume that customer behaviour patterns remain stable after model development.

Concept drift represents a major challenge because changes in data distribution can reduce prediction performance after deployment. Adaptive learning approaches provide promising solutions by enabling models to update continuously according to changing customer behaviour Gama et al. (2014); Bifet et al. (2009).

Real-time customer analytics and online learning frameworks should therefore become important components of future churn prediction systems to maintain model reliability in dynamic environments.

10.1.3. Limited Generalization Across Domains

Most churn prediction models are developed for specific industries, such as telecommunications, banking, or e-commerce. Although these models often achieve high performance within individual domains, their generalization capability across different industries remains limited.

Differences in customer behaviour, available features, business objectives, and regulatory environments create significant challenges for transferring churn models between domains.

Future research should investigate transfer learning, domain adaptation, and cross-domain learning approaches to

Table 6
Comparative Analysis of Customer Churn Prediction Studies

Reference	Domain	Data Source	Features	Methods	Limitations
Mettle et al. (2026)	Telecommunication	Telecom customer dataset	Customer demographics, service usage and account attributes	Comparative machine learning classification	Performance varies across classifiers; class imbalance remains challenging
Manzoor et al. (2026)	Telecommunications	IBM Telco and Maven Telecom datasets	Customer demographics, tenure, service usage, and billing information	e-Profits evaluation incorporating customer value, retention probability, and intervention costs	Evaluation limited to two telecom datasets; cross-industry generalization requires further validation
Barsotti et al. (2024)	Telecommunication	Large-scale telecom customer records	Usage behaviour, service interaction, customer activity	Survey of ML and DL churn techniques	Limited focus on profitability
Sikri et al. (2024)	Telecommunication	Telecom customer dataset	Customer demographics, service-related attributes, account information	Machine learning classification models with performance comparison	Handling imbalanced churn classes and improving prediction reliability
Ouf et al. (2024)	Telecommunication	Big data customer datasets	Usage behaviour, customer interaction history	Big data analytics and ML models	Data complexity and scalability issues
Adeniran et al. (2024)	Telecommunication	Telecom customer behavioural data	Call records, service usage, customer profile	Machine learning classification models	Limited explainability
Maan and Maan (2023)	Customer Analytics	Customer behavioural datasets	Customer attributes and behavioural indicators	Explainable machine learning	Explanation consistency challenges
Ahmad and Islam (2021)	SME Services	SME customer datasets	Transaction history, relationship duration	Traditional ML models	Limited data availability
De Caigny et al. (2020)	Banking	Customer interaction information	Customer behaviour and relationship data	Machine learning with additional information sources	Interpretability requirements
Ngai et al. (2009)	Customer Analytics / E-commerce	CRM and customer relationship datasets	Customer behaviour transaction information, interaction history	Data mining techniques and CRM analytics	Limited application of modern AI techniques

develop more flexible churn prediction frameworks capable of adapting to multiple industries.

10.1.4. Limited Integration of Causal Decision-Making

Traditional churn prediction models mainly focus on identifying customers with a high probability of leaving.

However, predictive accuracy alone does not answer the more practical business question:

"Which retention action will effectively prevent customers from leaving?"

Causal inference and uplift modelling provide promising solutions by estimating the effect of intervention strategies rather than only predicting churn probability. Unlike conventional prediction models, causal approaches attempt to identify whether a specific retention action changes customer behaviour.

Athey and Imbens (2016) introduced modern causal inference approaches for estimating heterogeneous treatment effects, enabling organizations to understand how different customers respond to specific interventions. Similarly, Künzel et al. (2019) proposed meta-learning approaches for estimating individualized treatment effects using machine learning methods.

Uplift modelling has also gained attention in customer analytics because it identifies customers who are most likely to respond positively to retention campaigns. Gutierrez and Gérardy (2017) reviewed uplift modelling approaches and highlighted their importance for optimizing targeted interventions.

Therefore, future churn prediction research should move beyond predictive analytics toward causal and prescriptive retention systems that determine not only who is likely to churn but also which actions can effectively improve customer retention.

10.1.5. Explainability and Trust Challenges

Advanced models such as deep neural networks and ensemble learning algorithms often provide improved predictive performance but operate as black-box systems.

Limited interpretability can reduce user confidence and restrict practical adoption, especially in high-impact domains such as banking, healthcare, and insurance.

Explainable AI approaches provide opportunities to improve transparency by identifying the factors influencing churn predictions. However, future research should focus not only on generating explanations but also on evaluating their reliability, stability, and usefulness for end users and decision-makers (Guidotti et al. (2019); Arrieta et al. (2020); Doshi-Velez and Kim (2017)).

10.1.6. Fairness and Ethical Decision-Making

Customer churn prediction systems rely on historical customer data, which may contain existing biases. Biased data collection, unequal representation, proxy variables, and historical business decisions can result in unfair prediction outcomes.

Although fairness-aware machine learning has received increasing attention, fairness remains an important consideration when churn predictions are used to allocate retention offers or other customer interventions.

Future churn prediction systems should incorporate fairness evaluation metrics and bias mitigation strategies

throughout the machine learning lifecycle (Mehrabi et al. (2021); Suresh and Gutttag (2021); Pessach and Shmueli (2022)).

10.1.7. Privacy and Data Security Challenges

Modern churn prediction systems require extensive customer information, including behavioural, transactional, and interaction data. However, increasing dependence on customer data creates significant privacy and security concerns.

Federated learning and privacy-preserving machine learning provide promising solutions by enabling collaborative model training without requiring raw customer data to be centrally shared.

Future research should investigate efficient federated churn prediction frameworks capable of addressing communication costs, non-independent and heterogeneous data distributions, client drift, privacy risks, and fairness challenges (McMahan et al. (2017); Li et al. (2020); Karimireddy et al. (2020)).

10.2. Future Research Directions

Future churn prediction research should move beyond conventional predictive modelling toward more intelligent, adaptive, and responsible approaches. The following directions highlight emerging opportunities for improving customer behaviour modelling, retention decision-making, and practical deployment.

10.2.1. Advanced Representation Learning for Customer Behaviour Modelling

Most existing churn prediction approaches represent customers using manually engineered features extracted from transactional, demographic, and behavioural data. However, complex customer ecosystems contain hidden relationships among customers, products, services, and interactions that may not be fully captured by conventional feature-based models.

Future research can explore advanced representation learning approaches, including graph-based learning, to model complex customer relationships and behavioural dependencies. Graph neural networks provide a potential direction by representing entities and their relationships as interconnected structures, enabling models to capture relational information beyond traditional tabular representations (Wu, Pan, Chen, Long, Zhang and Yu (2021); Zhou, Cui, Zhang, Yang, Liu, Wang, Li and Sun (2020)).

Although graph-based approaches have demonstrated success in graph representation learning and relational data analysis, their application in customer churn prediction remains relatively limited. Future studies should investigate how graph-based customer representations can be integrated with profit-driven evaluation, explainable AI, and fairness-aware retention strategies.

10.2.2. Multimodal Customer Intelligence

Future churn prediction systems should integrate multiple data sources, including transactional information, customer reviews, social interactions, service conversations, and behavioural sequences.

Multimodal learning approaches provide a more comprehensive understanding of customer behaviour by integrating heterogeneous data sources, including structured customer records, textual feedback, behavioural sequences, and interaction information. Combining multiple data modalities can improve churn prediction by capturing complex relationships between customer characteristics and dissatisfaction patterns.

Baltrusaitis et al. Baltrusaitis, Ahuja and Morency (2019) provided a comprehensive review of multimodal machine learning and highlighted its ability to integrate information from multiple modalities for more comprehensive machine learning systems. Future churn prediction systems can benefit from multimodal frameworks that combine transactional, behavioural, and unstructured customer information to achieve more accurate and context-aware retention decisions.

10.2.3. Reinforcement Learning for Autonomous Retention

Most existing models for customer attrition focus on predicting when a customer might leave than figuring out the best actions to take. Reinforcement learning offers a way to create strategies that can adapt over time by learning what actions work best based on how customers respond.

Future systems that help keep customers might use both predicting churn and reinforcement learning. This would allow them to suggest actions and keep improving based on what customers do.

Mnih et al. Mnih et al. (2015) demonstrated that deep reinforcement learning can be used to learn strategies in a sequence of sequential decisions. This sets the stage for using reinforcement learning to create strategies that're both adaptive and tailored to individual customers.

10.2.4. Towards Autonomous Intelligent Retention Ecosystems

The future of dealing with customer attrition is likely to move from models that only predict churn and instead create systems that are smart and run on their own.

These systems should be comprised of:

- Intelligence to estimate the risk of a customer leaving
- Causal inference to find the best actions to take
- Multi-stakeholder consultation to ensure the actions are consistent across the various stakeholders.
- A cost-benefit analysis to ensure that the actions are profitable to the business.
- Explainable AI – ensure decisions are understandable

- Fairness-aware learning, ensuring fairness in decisions.
- The privacy-preserving learning to protect customer information.
- Reinforcement learning to take actions that change over time.

So future research on predicting customer attrition should focus on creating intelligent systems that can manage prediction accuracy making money, fairness keeping data private and building long-term relationships, with customers.

11. Conclusion

Many things have been improved regarding customer churn prediction from mere statistics to more intelligent systems that can retain customers. This review looked at how the ways to predict customers leaving have developed. It presented machine learning, learning and how combining various models made AI more understandable, and how much money is made by keeping data safe and taking action to keep customers.

The study shows that even though new AI models are really good at predicting just being accurate isn't enough for business decisions. There are many ways that get that out of the way despite the fact that many ways fail to capture the value of a customer, if it is just, to keep data safe, make decisions clear and take action that it is worth the cost. In order to improve the future of predicting churn, it must change from predicting to helping make decisions that work.

This review argues that the emphasis on the profit-making aspect of the business, rather than considering customer value and business objectives, yields results. Being able to explain itself and have a fair treatment policy is also vital when it comes to decision-making transparency and equity. Ways to keep data safe especially using a method called learning help use customer data without giving away secrets.

In fact, as you can see from the studies, there are not many links between the various studies on predicting churn. Although some fields (such as telecom, banking, online shops, health care, subscription services) have methods, problems such as how people change and how well models work in different places are still not solved, and how to use models in real-life situations to make smart choices is not solved.

System design should not simply look for customers who could walk out the door, but determine what steps can actually help to keep customers. Graphs, such as being able to understand why things happen, being able to use things that help, learning from experience and making smart choices can help in creating systems that function independently and can be customized to each customer.

This review concludes by examining the evolution of customer loss prediction from a basic issue to a more complex problem. The next generation of considering customer

loss is about systems that are good at predicting customer churn are good at making money while also being good at keeping people happy, safe and loyal for a while.

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